

PARTICULAR SPECIFICATION P2

Preliminaries and General

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P2.1 INTRODUCTION

P2.1.1 Overview

- (a) This part of the Specifications covers the duties required of the Contractor to administer the Works and to coordinate with Other contractors through the S.O..
- (b) The Other contractors include Contract 1A (C1A), which includes various site enabling works, as further described herein, to support the Board and the S.O. and for use by all contractors appointed by the Board for the successful implementation of TWRP, Phase 1.
 - (i) Contract 4A Biosolids and Digestion. The Digestion Pre-treatment 1 Plant will be located in the Biosolid building's courtyard (also known as THP Courtyard) as indicated in the Drawings, within the vicinity of C4A.
- (c) The Contractor shall submit design proposals to the S.O. for all temporary works necessary for the construction of the Permanent Works as well as any elements of the Permanent Works detail designed by the Contractor. The Contractor shall be responsible for obtaining all necessary authority approvals for the elements of works designed by the Contractor for the temporary and permanent Works.
- (d) The Contractor shall provide the Qualified Person (QP) Design and QP Supervision for any structural and geotechnical aspects as applicable for all works and any temporary works and any elements within his scope of works unless specified in the Particular Specifications. The presence of the S.O. during the construction and commissioning shall not absolve the Contractor of his responsibility to ensure proper construction of the structural works and any temporary works and any elements of the Permanent Works detail designed by the Contractor.

P2.1.2 S.O.'s Approval

- (a) When the approval of S.O. is given for any element of the Works, such approval shall not absolve the Contractor from his liability to comply with the Contract and to make good any errors, damages, or defects which may be discovered or occur subsequently.

P2.1.3 Term Definitions

- (a) The following definitions apply to all items in the Specifications, Drawings, and Schedule of Prices:
 - (i) "Shall" – when used in these Documents implies mandatory action by the Contractor;
 - (ii) "May" – when used in these Documents implies optional action by the Contractor or by others;

- (iii) “Will” – when used in these Documents implies actions to be taken by others, unless otherwise apparent from the context in which it is used; and
- (iv) “Provide” – when used in these Documents in relation to MEICA, architectural and temporary civil and structural elements of the Works, this shall mean that the Contractor shall carry out the following activities to the extent required under the Contract:
 - 1) Design, supply, off-site inspection & testing, shipping & delivery to Site, off-loading, supervision, site installation, testing & commissioning, performance testing, provision of O&M information, as-built drawings/models, maintenance from installation to handover, rectification of defects along with any other specified requirements or activities necessary for the completion of the Works.
- (v) “Provide” – when used in these Documents in relation to permanent civil and structural elements of the Works, unless specified otherwise, this shall mean that the Contractor shall carry out the following activities to the extent required under the Contract:
 - 1) Design, supply, off-site inspection & testing, shipping & delivery to Site, off-loading, supervision, construction, testing & commissioning, performance testing, as-built drawings/models, maintenance from construction to handover, rectification of defects along with any other specified requirements or activities necessary for the completion of the Works.

P2.1.4 Insurances and Claims Procedure

- (a) The Contractor shall note that the Board has made special arrangements under the Contract for the following insurances:
 - (i) Contractor’s All Risks including Third Party Insurance; and
 - (ii) Excess Third Party Liability Insurance Policy.
- (b) In the event of any claims, the Contractor shall comply with the Claims Procedure for Contractors’ All Risks insurances included in Volume 1 of the Contract Documents.
- (c) In the event of any incident at the Site, the Contractor shall comply with the reporting requirements in the Specifications and as required in the claim’s procedures (if applicable).

P2.1.5 Claims for Damage to Persons or Property

- (a) Any claim received by the Board in respect of matters in which the Contractor is required under the Contract to indemnify the Board will be passed to the Contractor who shall likewise inform the Board of any such claim which is submitted directly to him by a claimant. The Contractor shall do everything necessary, including notifying the insurers of claims received, to ensure that all claims are settled properly and expeditiously and shall keep the Board informed as to the progress made towards settlement, failing which the Board shall be entitled to make direct payment to claimants of all outstanding amounts due to them in the Board's opinion and without prejudice to any other method of recovery to deduct by way of set-off the amounts so paid from sums due or which become due from the Board to the Contractor. If the Contractor receives a claim, which he considers to be in respect of matters for which he shall be indemnified by the Board under the Contract, he shall immediately pass such claim to the Board. All information as aforesaid shall be given in writing and shall be copied to the S.O..

P2.2 STANDARDS

- (a) The Contractor shall deliver the Works using the standards specified in the Contract. For more information regarding compliance with standards, refer to General Specification G2.4 Standards.

P2.3 SITE LOGISTICS

P2.3.1 Temporary Occupation Licences

- (a) The Contractor shall note that Temporary Occupation Licence (TOL) fees are not required for the Site.
- (b) The Contractor shall allow in his rates and prices for TOL fees for any other site required by the Contractor for his own use for the execution of the Works other than as provided in the Specifications.

P2.3.2 Project Occupier/ Field Mobilisation/ Occupier of Workplace

- (a) The Contractor shall note that there has been an update of requirements for registration of construction worksites with Ministry of Manpower (MOM) from 1 April 2019. The Contractor shall comply with the new requirements by MOM and Building and Construction Authority (BCA) for any works to be carried out at the Site. This does not affect existing WSH regulatory requirements on the worksites. Construction worksites will still need to comply with relevant WSH regulations, e.g. WSH (Construction) Regulations and WSH (Safety and Health Management System and Auditing) Regulations. The Contractor shall conduct and present External WSH audit report to the S.O. within two (2) months upon award of the Contract. After the Contract Award, the Contractor may start site preparations on Site. BCA's permit to carry out structural works is still required for building works as defined under the Building Control Act and its Regulations. The Contractor shall comply with the Building Control Act and its Regulations and relevant WSH regulations for the works to be carried out on site.

- (b) The Site will be under the charge of the Other contractors. The Contractor shall coordinate with Other contractors to ensure the handover of the Site. The Contractor is to document the handover of the Site from Other contractors. The document should minimally indicate when the area is handed over, the extent of the area handed over and endorsement/acknowledgement by both parties with date specified. Once handed over the Site will no longer be under the Other contractors' charge.
- (c) The Contractor shall note the progressive transfer of work areas within the Site to Other contractors. The S.O. will facilitate the handover of the specified work area from the Contractor to the Other contractors. The Contractor shall coordinate with the S.O. and the Other contractors to ensure the handover of the work area. The Contractor is to document the handover of the specified work area. The document should minimally indicate when the area is handed over, the extent of the area handed over and endorsement/acknowledgement by both parties with date specified. Once handed over, the work area within the Site will no longer be under the Contractor's charge.
- (d) The Contractor may, with the agreement of the S.O., vacate and hand back any Staging Area and any temporary work areas. The Contractor is to document the handover of the areas previously under their responsibility to the Other contractors. The document should minimally indicate when the area is handed over, the extent of the area handed over and endorsement/ acknowledgement by both parties with date specified.
- (e) The Contractor shall comply with the requirements defined by the C1A contractor, subject to S.O.'s approval, for any work to be carried out by the Contractor in the Common Areas of the TWRP Area managed by the C1A contractor.
- (a) The Contractor shall comply with the requirements defined by the C4A contractor, subject to S.O.'s approval, for any work to be carried out by the Contractor in the TWRP Biosolids & Digesters area managed by the C4A contractor.
- (f) The Contractor shall comply with the requirements defined by Other contractors, subject to the S.O.'s approval, for any work to be carried out by the Contractor in the Other contractors' site.
- (g) The Contractor shall, subject to the S.O.'s approval, define reasonable requirements for any works on its Site to be done by Other contractors and shall coordinate with such Other contractors for such works including compliance with the contractor's requirements.
- (h) Aside from complying with all statutory requirements, the Contractor shall comply with the Health, Safety and Environmental requirements in the Contract.

P2.3.3 Surveying and Setting out the Works

- (a) The C1A contractor will survey, mark and peg the boundaries of the TWRP Area, including all interface points between the C1A contractor, the Contractor and Other contractors. The Contractor shall verify the survey carried out by the C1A contractor for the Site and shall coordinate with the C1A contractor and S.O. to resolve any discrepancies in surveys. Once the survey is confirmed, the Contractor shall be fully responsible for the Site and for all surveys and setting out of the Works within the Site. The Contractor shall employ a Registered Surveyor for all survey works including coordination with the C1A contractor survey.
- (b) The C1A contractor will provide services to coordinate survey control of the TWRP Area, including base survey control and verifying survey by the Contractor of the Site. The objective of these services is to assist the S.O. in verifying proper survey and coordinate control for all the Works for the Site. The Contractor shall fully cooperate with C1A survey services. At the S.O.'s discretion, he may direct the C1A contractor to survey any area within the Site as necessary and the Contractor shall cooperate with such survey.
- (c) The Contractor shall verify any survey carried out by Other contractors for the Works and shall coordinate with the Other contractors and S.O. to resolve any discrepancies in surveys. Once the survey is confirmed, the Contractor shall be fully responsible for the Works and for all surveys and setting out of the Works within the Site. The Contractor shall employ a Registered Surveyor for all survey works including coordination with the Other contractors' survey.
- (d) The co-ordinates, levels and dimensions shown in the Drawings or given in the Specifications, are based on the SVY21 plane coordinate system and Singapore Height Datum. The Contractor shall coordinate with the S.O. and verify the datum system prior to starting construction.
- (e) The Contractor shall carry out topographical surveys of the Site prior to the Works to verify any existing levels for the certification of work done.
- (f) The C1A contractor will establish and maintain eight (8) permanent survey control markers in the SVY21 plane coordinate system, with elevation referenced to 0m above the Singapore Height Datum (SHD) in the TWRP Area. The markers will be used for the setting out of all facilities by all TWRP contractors including the Contractor.
- (g) The Contractor shall coordinate with all interfacing contractors to ensure that the coordinates for their interface locations match. Interfacing contractors include but not limited to the following; C1A, C2A, C3A, C3C, C3D1, C3D2, C3B1, C3B2, C4A, C5A, C6A and C6B contractors. If there is any discrepancy between the Contractor survey and the survey of the interfacing contractor, the two contractors shall employ their Registered Surveyors to conduct a joint survey and resolve the discrepancy.

P2.3.4 Other Surveys and Investigations**(a) Field Investigations**

- (i) Prior to the excavation at Site, the Contractor shall carry out field investigations to ascertain the ground conditions. The Contractor shall submit the proposed field investigation plan indicating the type of investigation, location, extent and any other details to the S.O. for approval prior to carrying out any field investigations.
- (ii) The Contractor shall submit the results of the field investigations to the S.O. prior to the commencement of work. The Contractor shall update the services drawings and model to reflect the results of the investigation.
- (iii) Field investigation shall include but not be limited to the following:
 - 1) Soil investigation survey. The Contractor may, at his own cost, carry out additional soil investigation survey to obtain necessary soil and ground water data and information as required for the design of foundations and reinforced concrete works, where deemed necessary by the Contractor; and
 - 2) Precondition and Existing services surveys. The Contractor shall clearly mark the location of existing services on the topographical survey drawing and peg on Site. The Contractor shall note the location and perform pre-condition surveys of all nearby existing structures, facilities, utilities and roads that may be within the zone of influence of its Works.
- (iv) The field investigations shall include confirmation of position and level information included on the Drawings and determination of the presence, dimensions and level of all utilities, pipelines, drains, cables, chambers, duct banks, cable pits, manholes and structures and the like, which are related to or may be affected by the Works. If the field investigations reveal any significant discrepancies from the Drawings, the Contractor shall inform the S.O. immediately of all such discrepancies. At all times the Contractor shall work to the instructions of the S.O. regarding any modification to the design of the Works made necessary by the identification of such discrepancies.

(b) Surveys during Construction

- (i) The Contractor shall carry out the following surveys during construction and submit to the S.O. for approval:
 - 1) Construction Noise Measurement. As specified in the HSE Specifications;
 - 2) Photo survey. Monthly photo survey of the Site;

- 3) Time lapse surveys. The Contractor shall provide a time lapse video camera (minimum 4K video quality). Time lapse survey shall comply with the following minimum requirements:
 - Camera(s) to be positioned at external vantage location(s) to be proposed by the Contractor and approved by the S.O. to provide coverage of the Site;
 - The system shall capture and upload images every 10 minutes, 24 hours per day;
 - The Contractor shall ensure that the cameras are suitable for outdoor use and account for exposure to the elements. The Contractor to provide for all necessary poles for mounting equipment, supports, cabling and foundations as required;
 - The time lapse cameras shall be installed within three (3) months upon award and shall remain in place until the Works is substantially complete; and
 - The Contractor shall service and maintain the cameras and shall provide replacement units as necessary to ensure continuous filming and back-up the recorded data. The recorded, unedited time lapse footage for each location shall be provided to Board and S.O. monthly. The Contractor shall engage a professional editor to produce a full-time lapse videos (one for each of the location) for the Contract period. It shall be submitted to the Board and S.O. at the end of the Contract.
- 4) Levels survey. Progressive level survey for each area of work in accordance with the Programme for the Works; and
- 5) Laser Scanning. Provision of point cloud with BIM rendered progress files for all elements of the Works during the month for comparison with the project BIM model. Laser Scanning shall be completed every six (6) months. Laser scanning shall comply with the following minimum requirements:
 - Scan formats to be in PTS; E57 or POD;
 - Scans shall be performed at a minimum resolution of 6mm @ 10m;
 - Contractor to confirm that all scan data is in the correct coordinate system and has been properly referenced to the BIM;
 - Scanning equipment shall be RTC360, P40, ZF5010/HDS6000, or equivalent;

- Scanner Precision should be 1mm + 10ppm or better;
- Relative accuracy of least squares adjustment should be better than +/- 5mm;
- Scans submission to the S.O. are to ensure that all Works including walls, cast-in items, equipment, valves, piping and instrument locations are verified and shown overlain on the updated BIM model;
- Contractor to provide a 12 bay Network Accessible Server (NAS) with total raw storage capacity of 96TB for the S.O.'s storage of scan and BIM rendered files. The Contractor shall provide warranty and maintenance for the NAS for the life of the Contract including if and when necessary upgrades and replacement of the NAS; and
- Contractor to provide a 2TB portable hard disk for transfer of scan files to the S.O.. Contractor shall ensure that the disk is scanned and clear of virus prior to transferring files to the S.O..

(c) Post-Construction Surveys

(i) The Contractor shall carry out the following post-construction surveys:

- 1) **As-built topographical survey.** The Contractor shall carry out an as-built topographical survey for the completed Works within the Site. The survey shall verify ground levels and locations, size and levels of all structures, buildings and other surface features. As-built topographical survey shall be endorsed by a competent Registered Surveyor holding a valid practicing certificate issued by the Land Surveyors Board, Singapore. The topographical survey shall comply with the requirements as specified in General Specification G2.5 Condition Surveys.
- 2) **Post-construction survey.** The Contractor shall, upon completion of the Works, engage qualified firm/personnel to carry out post construction survey for the Site for all existing properties, buildings and structures, LTA Structures and road elements, and other adjacent structures within the influence zone from the contract boundary together with any additional properties affected by the construction works, as stipulated in the latest BCA guidelines and as specified. The survey shall cover the structural condition of all existing structure (including those under construction or completed by Other contractors during the duration at the point of survey or at other points of time as the S.O. deems necessary) recorded in the pre-construction survey. The aim of the survey will be to identify any deterioration in the structure condition that cannot be attributed to normal operational wear. The post-condition survey shall comply with the requirements in General Specification G2.5 Condition Surveys.

P2.3.5 Special Hazards and Unsafe Working Conditions

- (a) The Board and/or Other contractors may store hazardous substances and dangerous goods at various locations in the TWRP Area, as stipulated in the WSH Act and guidelines, during the testing, commissioning and operations periods of TWRP. The Contractor shall develop an emergency plan of action for any special hazards within their Works to contribute to the site emergency plan. The Contractor shall take note of the emergency plan of action to be taken in the event of a major leak, fire or explosion and shall follow any instructions given by the S.O. and/or Other contractors in the event of an emergency. The Contractor shall also co-operate, when required, in drills designed to test the efficacy of the plan.

P2.3.6 Site Constraints

- (a) The Contractor shall note the constraints in and around TVB, the TWRP Area and the Site for delivery of materials and equipment and make appropriate allowances for this in his rates and prices.
- (b) The Contractor shall note that there will be various construction contracts on-going during the Contract period. The Contractor shall liaise closely with the S.O. and Other contractors and interfacing contractors on the coordination of the Works or the interfacing between the various parties working within and external to the Site.
- (c) The Contractor shall note that various parcels of land have been allocated to different contractors for their site offices and staging areas and these contracts will be mobilised at different times during the Contract period. TWRP contractors will establish various temporary access ways to the TWRP Area roads provided by the C1A contractor.
- (d) The C1A contractor will develop and manage a Traffic Management Plan (TMP) for use of the roads on the TWRP Area and access via Routes 2 and 3 to be approved by the S.O.. The TMP will define traffic rules and procedures for the TWRP Area and the Contractor shall fully comply with all such rules and procedures, which may change during the Contract period.
- (e) The Contractor shall note that NEA will be engaging contractor(s) for the design and construction of the IWMF during the Contract period. The IWMF is expected to be in operation late in the Contract period and the C1A TMP will take into account the NEA contractors.
- (f) The Contractor shall note that PUB (WSN) will be engaging a contractor for the design and construction of the NEWater and Industrial Water delivery network. The WSN contractor is expected to be constructing these works during the Contract Period. The C1A TMP will take into account the PUB (WSN) contractor.
- (g) The C1A contractor will establish an Access Control System (ACS) for managing access and security to the TWRP Area and North Staging Ground (NSG). Refer to Site Access in P2.5.3.

- (h) The Contractor shall note that the working areas in the sites of Other contractors for the TWRP project will be shared working space with the respective contractors and will only be available to the Contractor at different phases and timing within the Time for Completion. The Contractor shall coordinate with the Other contractors on the access to their site and possession of part of the site required for the Works, and schedule the installation, testing and commissioning to achieve the milestones required.
- (i) The Contractor shall note that there are dormitories operated by third parties near the Site. The Contractor shall comply with and adopt noise mitigation measures where necessary in accordance with the Environmental Pollution Control Act, Control of Noise at Construction Site Regulations and as specified in the HSE Specifications.
- (j) The Contractor shall use all reasonable means to prevent noise and annoyance to the inhabitants of the area in which the Works is situated, and all machinery must be of such design and so arranged as to be reasonably free from noise in operation. Where air compressors or pneumatic tools are being used in the Works, the Contractor shall arrange, as required by the S.O. for such compressors and tools to be fitted with silencers of approved design or he shall adopt other approved means of reducing the noise. The Contractor shall maintain and operate all fuel burning plant and equipment such that the requirements of local law and regulations are complied with. No claim by the Contractor for complying with the provisions of this Clause shall be entertained by the Board.

P2.4 TEMPORARY WORKING AREAS AND STAGING AREAS

P2.4.1 Overview

- (a) The Contractor will be allocated a staging area at TWRP Area for the Contractor's use as shown in the Drawings. It is envisaged that the Contractor's site office will be located at the staging area. The staging area will be surveyed and pegged by the C1A contractor.
- (b) The staging area will be at elevation + 5m SHD, level and compacted. The perimeter of the staging area may be sloped to the TWRP Area roads. The Contractor shall conduct a joint inspection of his staging area with the S.O. and C1A contractor prior to commencement of any work in the staging area to confirm the condition and survey. Any discrepancies shall be finally decided by the S.O..
- (c) The Contractor shall submit the staging areas hoarding alignment with coordinates on topographical survey drawings for the S.O.'s approval prior to setting up at the staging areas.
- (d) Temporary roads on the TWRP Area will be available to the Contractor for access to the Site including staging and work areas. The Contractor shall be responsible for the establishment of roads within the Contractor's staging areas and temporary work areas and for road access between its staging area and working areas to the TWRP Area roads.

P2.4.2 Use of Temporary Working and Staging Areas

- (a) The Contractor shall note that his staging and work areas may be at different elevations compared to the TWRP Area roads. The Contractor shall allow for this in establishing access into its work area within its rates and prices. The approximate levels during initial handover of the site to the Contractor are +5m SHD for C4B's staging area.
- (b) The Contractor shall provide approved project signboards, hoarding with contract code (C4B) in large capital letters painted at regular intervals of 20m and gates for his staging area.
- (c) The Contractor shall ensure that the staging area and temporary working areas are used solely in relation to the delivery of the Works.
- (d) The Contractor shall provide such temporary offices, buildings, tanks, etc. as may be necessary and proper for his general use in connection with the Works, and for the use of persons employed by him. The nature of the offices, buildings, tanks, etc. and the positioning of them shall be subject to the prior approval of the S.O..
- (e) The Contractor shall maintain all offices, buildings, tanks, etc. referred to in this and succeeding clauses in good condition and comply with all the requirements of the Board and S.O..
- (f) The Contractor's temporary working areas will need to move occasionally based on the Work being carried out at any time. The extent of each temporary working area and the period for occupation shall be such as the S.O. considers necessary considering the Contractor's reasonable requirements which shall be submitted to the S.O. in the form of a method statement as soon as practicable after commencement of the Works on the Site and having regard to the Contractor's programme.
- (g) The Contractor's temporary working areas shall be hoarded prior to the commencement of works at any area. The Contractor shall submit the proposed working area hoarding plan with coordinates on the topographical survey drawings for the S.O.'s approval prior to setting up at each temporary working area.
- (h) The Contractor shall allow for all costs associated with the erection and dismantling of the hoarding upon handover of the facilities to the S.O..

P2.4.3 Additional Temporary Staging and Storage Areas

- (a) The Contractor shall note that a 3.5 ha area along Tuas South Way and Tuas South Street 15 referred to as South Staging Ground 1 (SSG1) has been secured for use by the Board as a temporary staging or storage area.

- (b) SSG1 may be available to the Contractor (and Other contractors) as a temporary staging or storage area on an as-needed basis up to end 2023 subject to approval by the S.O.. There will be no TOL fees required from the Contractor for the use of SSG1, but there will be no guarantee that SSG1 will be available to the Contractor for the areas and/or time periods requested by the Contractor, as it is subject to S.O.'s approval. The Contractor shall be responsible for the following:
- (i) Obtaining all approvals for road openings and providing access into the site;
 - (ii) All site preparation and maintenance works, including but not limited to:
 - 1) levelling of ground to the desired platform levels, compaction (where required); and
 - 2) hoarding, lighting, drainage, erosion control, vector control, housekeeping, grass cutting and other facilities and activities needed to maintain health, safety, security and environment of its allocated area and staff working within the allocated area to the requirements of the prevailing laws and regulations and Specifications.
 - (iii) Traffic management and ensuring that adequate washing bay facilities are provided to ensure that vehicles are cleaned prior to entering public roads;
 - (iv) Any infrastructure required for any use of the area in compliance with requirements of authorities;
 - (v) Maintenance of common areas shared with Other contractors, including but not limited to levelling of ground, removal of debris and carrying out of vector control, grass cutting and housekeeping; and
 - (vi) Provision of utilities (water, power, sanitation) for their own usage, including all consultation with authorities, connection fees and usage charges, and subsequent removal and reinstatement.
- (c) The Contractor shall submit its plan of use for any staging areas to the S.O. for approval. The Contractor shall note that the staging grounds will also be utilised and occupied by Other contractors. The Contractor shall carry out all coordination, interfacing with Other contractors and provide the resources to carry out appropriate maintenance of common areas shared with Other contractors. Common areas include but are not limited to common accesses and entrances. Maintenance of common areas may include the levelling of ground, removal of debris and carrying out of vector control, grass cutting and housekeeping along the common access road. After use, the Contractor shall restore any areas of the staging grounds used to the pre-use condition and shall return the area as required by the authorities. Pre-and post-condition surveys shall be completed by the Contractor in coordination with the S.O and submitted to the S.O for approval.

- (d) The Contractor shall note that should he require additional areas for his plant, equipment and materials other than at the Tuas WRP area and SSG1 areas, he shall be entirely responsible for making the necessary arrangements for hoarding, protection, maintenance, access, etc. subject to the approval of the S.O.. TOLs, rental charges and all costs arising thereof shall be borne by the Contractor. The Contractor shall reinstate the lands affected to the satisfaction of the S.O. on completion of the Works.

P2.4.4 Reinstatement of Working Areas and Staging Areas at Site

- (a) The Contractor shall reinstate the staging areas and any temporary working areas allocated to the Contractor as soon as possible after all work in those areas have been completed to keep the period of occupation to a minimum. The Contractor shall return the staging and temporary working areas to the S.O. in the condition received upon completion of its use of the staging area. The reinstatement shall include removal of all surface and subsurface facilities and infrastructure installed, clean-up of any waste disposed, removal of vegetation, levelling and compacting of ground to elevation of +5m SHD.

P2.4.5 Washing Facilities

- (a) The Contractor shall note that all roads within the Site shall be regarded as public roads with respect to the level of cleanliness and upkeep.
- (b) The Contractor shall provide, operate and maintain washing bays or an equivalent washing service (subject to S.O. approval) to ensure all the Contractor's vehicles leaving the Site during the Contract period are clean and vehicles wheels are generally free from debris. Washing services shall be operated during all working hours.
- (c) The Contractor shall design the washing facility to handle the washing of all types and numbers of vehicles used by the Contractor. The vehicle washing facility shall be of adequate width and length such that it completes a full rotation of wheel washing on both sides of the vehicle. The vehicle washing point shall be located such that vehicle wheels are cleaned the moment it travels on public road and no overflowing of water from the washing point to the public road shall be allowed. The wheel washing facilities shall also have adequate drainage and appropriate measure to prevent standing water and breeding of mosquito.
- (d) The Contractor shall provide the required number of workers, supervisors, equipment and machinery, water supply, silt trap and filtration system for recycling of the water etc. in support of the washing facility operations as required.

P2.4.6 Site Layout Plan

- (a) The Contractor shall establish a Site layout plan to ensure that the Site is designed to maximise the efficiency of operations to promote worker safety and productivity. The Site layout plan shall depict the proposed locations of temporary facilities within the construction site boundary. Temporary Facilities are defined as those facilities and areas delineated to specific tasks that support the construction process such as site office and washrooms, unloading areas, material storage/staging areas, machinery parking areas, fabrication yard, work areas, tool and equipment areas, washing points, vehicular access routes, and pedestrian walkways.
- (b) The required temporary facilities must be clearly identified on the plan. The size and location of the areas allocated for the temporary facilities must have thorough consideration of the project needs, site and surrounding conditions and the applicable regulations to mitigate the safety and health risks. The temporary facilities should be clearly demarcated with signage and barricaded where possible.
- (c) As a minimum, the temporary facilities should include the following:
 - (i) Separate site access routes should be provided for vehicular and human traffic to prevent any accidents to pedestrians, equipment, etc. within and around the Site. The perimeter of the Site traffic access route should be barricaded or fenced to prevent unauthorised entry. Entry and drop off points(s) should be created to facilitate effective control of loading/unloading goods and personnel movement within the Site area. Traffic access shall be wide enough for safety where separation is not practicable.
 - (ii) Access for pedestrian and vehicle shall be well illuminated and have firm surfaces with suitable width, adequate drainage, of adequate strength and gradient to allow safe movements of traffic. They shall also be free from obstruction and workplace hazards such as falling materials, slippery surface and tripping hazards.
 - (iii) Appropriate signage shall be put up to alert drivers and pedestrians of the speed limit, sharp bends, pedestrian crossing, vehicular movement, steep slopes, narrow roads, winding roads and/or slippery roads.
 - (iv) Proper locations shall be identified for material storage on Site. The amount and type of materials to be stored, loading and unloading area, the level of security and protection against weather, where necessary, shall be indicated on the layout plan. The storage area shall have sufficient working access, be well illuminated and clear of any tripping hazards.

- (v) Hazardous materials that are explosive, flammable, or toxic substances shall be properly stored and separated to prevent accidental combinations of materials that can create hazardous conditions and exposing site occupants to safety and health risks. In this respect, the requirements of National Environment Agency (NEA) on pollution control and Workplace Safety and Health Guidelines for hazardous materials storage shall be given due consideration when pricing for the project.
- (vi) The route for temporary services shall be carefully planned out to avoid unnecessary excavation and diversion as construction work progresses. The services shall be installed early and have provisions for temporary connections for site use to be made during the construction period where practicable.
- (vii) The Contractor's site office shall not be located too near to noisy activities such as mixer or located too near to Site roads in dusty conditions, or too remote with insufficient overview of the Site. The site office shall have adequate toilets, lightings and ventilation. Sufficient emergency exits, and fire extinguishers shall be provided in the site office. Access leading to the site office shall be well illuminated.
- (viii) Loading and unloading areas shall have adequate space on a firm and reasonably level surface for lifting, filling, parking and manoeuvrings.
- (ix) Rest areas, meals area, drinking water points, prayer room and adequate toilets shall be provided and indicated on the layout plan. These areas should be well illuminated, provided with adequate storage for it to be free from cluttering, free from dust, safely located away from vehicle movement as well as hazardous area and adequately ventilated. Sufficient emergency exits, and fire extinguishers shall be provided in these areas if its size requires. If these facilities are exposed to vehicular movement, rigid crash barriers or posts are to be installed.
- (x) Machinery parking areas shall be indicated on the layout plan. Designated machinery parking areas must be located on firm surfaces of reasonable level with appropriate bearing capacity to prevent any collapse of machinery. The parking area must be of suitable safe size, profile, and gradient with adequate turning radius for safe machinery movement and parking.
- (xi) The machinery parking area shall also have adequate drainage and be protected against unauthorised access, weather and accidental damage. Drip pans/drainage shall be appropriately used to catch hydraulic fluids and prevent slip hazards. Signage shall be put up to alert personnel of surrounding hazards and for observation of personal safety.

- (xii) Any fabrication yard shall be of adequate size for the type of fabrication work it is catered for. There shall be a well illuminated and clearly demarcated loading and unloading area, walkway of adequate width free of tripping hazards within the fabrication yard to facilitate safe movement of workers within the compound. Fabrication works involving hot works shall be separated from other fabrication yard with flammable material at a safe distance to prevent any hazardous condition and exposing site occupants to safety and health risks.
- (xiii) Separate fabrication yards shall be clearly demarcated with barricades or fencing, and appropriate signage shall be put up to alert personnel of surrounding hazards and to keep out unauthorised personnel access.
- (xiv) An adequately spaced area at a safely appropriate location shall be identified and indicated on the layout plan for safe and proper storage of tools and equipment on Site. The tool and equipment area shall have a clearly demarcated walkway of adequate width for proper storing and retrieving of tools and equipment, shall be well illuminated and shall be clear of any tripping hazards.
- (xv) The tool and equipment area shall also be protected against unauthorised access, weather and accidental damage. Where practical, CCTV shall be installed to cover the entry of the area to deter and prevent loss of tools and equipment.
- (xvi) An adequately spaced area at a safe appropriate location shall be identified and indicated on the layout plan for safe and proper storage of any free - issued equipment from Other contractors (supplying contractor) or delivered equipment from his sub-contractors on Site. The Contractor may also consider off-site storage at his own cost. The storage shall provide adequate storage conditions in accordance with the supplying contractors instructions and/or Specifications.
- (xvii) The Contractor shall provide a multi-purpose space within their temporary facilities to be used for other purposes in normal circumstances (as required by the Contractor) but which can be easily and quickly converted into Temporary Living Quarters (TLQ) for up to ten (10) pax upon instruction by the S.O.. The Contractor shall carry out all necessary application to the Ministry of Manpower (MOM) and any other required authorities. The TLQ shall include but not be limited to the following:
 - 1) Space to meet the minimum requirement for liveable space per occupant (min. 6m²), bed spacing, maximum occupancy per room; and
 - 2) Common toilets and showers (on the same floor as the TLQ) to meet the worker ratio requirement.

- (xviii) Only upon instruction by the S.O. to do the conversion the TLQ shall then be equipped to comply with all requirements issued by authorities including but not limited to Ministry of Manpower (MOM) and Building and Construction Authority (BCA).
- (d) The Contractor shall submit the Site layout plan for the Contract to the S.O. no later than 10 (ten) days before commencing any work on Site. The relevant safety requirements and considerations for the layout, planning and organisation of the Site shall be described in detail. Special risks involving specialised equipment and hazardous materials are also to be highlighted.
- (e) The Contractor shall resubmit for approval the updated Site layout plan whenever conditions in the current approved plan have changed, or the locations of temporary facilities required are altered over the duration of the project to suit the various stages of construction according to the Programme for the Works.

P2.5 SITE FACILITIES AND SERVICES

P2.5.1 Site Facilities

- (a) The Contractor shall provide and maintain a furnished site office in the designated staging area sufficient for managing the Part 1 Works.
- (b) The site office shall include rooms for meetings with the Board and S.O. during the Contract period.
- (c) The Contractor shall be responsible for the cleaning of the S.O. allocated area within the site offices.
- (d) The Contractor shall establish a guest Wi-fi system for the use of the S.O. and the Board within the site offices.
- (e) The Contractor shall provide and maintain a furnished site office for his personnel in a designated staging area (to be confirmed by the S.O. at a later stage) during the OM Period.
- (f) The Contractor shall provide for any temporary secure and lockable storage facility for any Spares as provided for under the Contract in Part 1. The Spares shall be stored as per the manufacturer's recommendations. The Contractor shall provide for the transfer of Spares from any temporary storage facility to the permanent storage area on the TWRP Area at the end of the Part 1 Contract (as confirmed by the S.O..)

P2.5.2 Site Canteen

- (a) The C1A contractor will operate a site canteen which will be available to the Contractor and all Other contractors. The canteen will serve three (3) types of meals; Malay food (Halal), Western food (Halal) and Chinese food (non-Halal). There will also be a drinks stall. The site canteen will operate daily (Monday to Saturday) from 07:00 to 19:00. The site canteen will be available by the Contract commencement date.
- (b) No other site canteens shall be permitted.

P2.5.3 Site Access

- (a) Access to the Site shall be through Routes 2 and 3 as shown on the Drawings. The C1A contractor will operate and maintain the access through Routes 2 and 3. Access will be available up to 24 hours per day 7 days per week as required to support the needs of the TWRP contractors. At various times access may be restricted or unavailable at either Route 2 or Route 3 due to various circumstances. At all times, other than emergency situations, access will be available at a minimum through one of the two access points.
- (b) The C1A contractor will develop, operate and maintain an Access Control System (ACS) to control the access and security of all persons and vehicles entering the TWRP Area. The Contractor is required to comply with the procedures for the ACS as approved by the S.O.. Key elements of the ACS will include:
 - (i) All persons and vehicles must be authorised to enter the TWRP Area;
 - (ii) All persons authorised to enter the site must have completed a safety induction course to be implemented by the C1A contractor. Details of the safety induction course are provided in Particular Specification P9 Health Safety and Environmental Management;
 - (iii) Full time workers and staff will be authorised through an electronic system including facial recognition and photo id cards. The system will require various information for each authorised worker and staff. The system will be used to prepare productivity and any other reports required by BCA;
 - (iv) Visitors and delivery drivers will be authorised through an electronic on-line advance registration system;
 - (v) Workers are required to be transported into the site on buses. Personal vehicles for Contractor's staff will be restricted to a limited number and will be authorised by the S.O.. Parking for Contractor's vehicles shall be within the Site;
 - (vi) Deliveries will be authorised in advance through an electronic online advance registration system;

- (vii) Motorcycles, bicycles, motorised scooters and all similar vehicles will be prohibited from entering the TVB. Motorcycle and bicycle parking may be available at the North Staging Ground; and
- (viii) Pedestrians who are authorised to enter the TVB will be allowed to enter through walkways on Routes 2 and 3. Pedestrian footpaths will be available.

P2.5.4 Site Security

- (a) The Contractor shall be responsible for providing security for his Site.
- (b) The C1A contractor will provide security for TVB and the TWRP Area. Security will include:
 - (i) Hoarding around the TWRP Area of the TVB;
 - (ii) CCTV Systems;
 - (iii) Access Control System;
 - (iv) Guards and sentry posts at Routes 2 and 3 access points; and
 - (v) Guard patrols of the perimeter of the TWRP Area.

P2.5.5 Roads and Site Transportation

- (a) The C1A contractor will operate and maintain an internal road network on the TVB to provide access to each of the TWRP contractor staging and work areas.
- (b) The internal road network will generally consist of paved roads meeting LTA type 2 standards. This may vary during the Contract period due to the requirements of the works of the C1A contractor and Other contractors. Levels, alignment and locations of the internal roads are subject to change. The Contractor will at all times be provided road access to the staging and work areas.
- (c) The C1A contractor, subject to the approval of the S.O., will implement a traffic management system for all roads in the TWRP Area. The system will include facilities and procedures to manage and control vehicular traffic, including signage, speed bumps, speed limits, traffic speed cameras. The Contractor shall comply with all traffic management requirements and shall take note that compliance with speed limits is one of the DTSS2 Life Saving Rules, which will be strictly enforced.

- (d) To the extent possible, the Contractor and Other contractors will be allowed to use as many authorised vehicles as required for the Work. However, due to the nature of TVB, the TWRP Area and access points, the volume of construction on TWRP Area, and the presence of other construction projects including IWMF and the Neste expansion program, traffic volumes may reach a level requiring implementation of restrictions on vehicle entry and movement on site. There may also be traffic congestion and delays on the surrounding roads leading to the Route 2 and 3 access points, congestion and delays on the Route 2 and 3 bridges and at the security access points, congestion and delays on the TWRP Area roads. The S.O. may implement various measures to manage traffic and congestion which measure may include but not necessarily be limited to the following:
- (i) Requirements for Contractor and Other contractors to provide forecasting of their traffic volumes and deliveries;
 - (ii) A system of coordinating and scheduling deliveries by the Contractor and Other contractors that may include limiting volumes of deliveries and vehicle movements at certain times;
 - (iii) Requirements for deliveries by Contractor and Other contractors to make deliveries during off-peak hours;
 - (iv) Requirements for Contractor and Other contractors to stage vehicles and deliveries in off-site locations;
 - (v) Compliance with changes in the traffic management plans to be managed by the C1A contractor under supervision of the S.O.;
 - (vi) Reductions in the number of personal vehicles that can be brought on TVB by the Contractor staff; and
 - (vii) A vehicle delivery logistics management system to be implemented by the C1A contractor, under the authority of the S.O.. The Contractor shall consider this possibility in planning and shall comply with all measures implemented to manage the volume and movement of traffic on site.
- (e) The Contractor shall take into account these potential traffic congestion and delays and measures by the S.O. in its works planning, programming and in its tender rates and prices.

P2.5.6 RORO Landing Facility

- (a) RORO (Roll-On Roll-Off) landing facilities will be operated and maintained by the C1A contractor in the TWRP Area as shown on the Drawings.

- (b) The RORO landing facility is for the use of all TWRP contractors, subject to the procedures and restrictions approved by the S.O.. A RORO landing facility in TWRP Area will be available for the duration of the Contract period.
- (c) The S.O. will work with the C1A contractor to issue procedures for use of the RORO landing facility.

P2.5.7 Transport and Logistics

- (a) The Contractor shall ensure that drivers comply with the traffic regulations as set out in the traffic management plan including speed limits. The Contractor shall ensure that passengers board and alight vehicles at designated points.
- (b) The Contractor shall transport his workers to and from the Site safely each day by bus and return them to off-Site dormitories. This requirement shall also apply to the workers under his sub-contractors. Under no circumstances shall the Contractor transport workers and personnel in open trucks. All costs for the transportation of the Contractor and subcontractor's workers and personnel shall be included in the rates and prices.

P2.5.8 Workers Dormitories to be outside TVB

- (a) In general, the Contractor shall not set-up any workers' quarters on Site, in the TWRP Area or on TVB unless otherwise specified. Instead, all workers shall be accommodated in external dormitories and transported to Site at the start of each working day and away from Site at the end of each working day. The Contractor shall provide suitable transportation for this purpose as specified in Particular Specification P2.5.7 and elsewhere in the Specifications.
- (b) The Contractor is encouraged to house the workers for the Contract in dormitories within the vicinity of TVB. The Contractor shall note that there is an available dormitory in Tuas by PUB. **Error! Reference source not found.** provides information on the dormitory.

Table P2-1: PUB Foreign Worker Dormitory in Tuas

S/N	Name of Dormitory	Name of Operator	Address	Telephone
1	Tuas South Lodge	Arina International Housing Pte. Ltd.	Tuas South Street 15	+65 9479 4372

P2.5.9 TWRP Area Lightning and Emergency Warning System

- (a) The C1A contractor will install and maintain a TWRP Area lightning and emergency warning system to inform all contractors working in the TWRP Area of potential lightning storms and other emergency events. The Contractor will be responsible for taking all appropriate actions in the event of lightning warnings or emergency events. All costs are deemed to be included in the tender sum.
- (b) The system will utilise strobe lights and sirens to warn all persons of forecasted lightning event and other emergency.
- (c) The Contractor will liaise with the C1A contractor on the procedures for implementation of the lightning and emergency warning system.

P2.5.10 Radio Communications System

- (a) The C1A contractor will provide a radio communications system for the use of the Board and the S.O. inclusive of the necessary operating frequencies and licence from IMDA.
- (b) The Contractor shall be responsible for providing their own radio communications system inclusive of the necessary operating frequencies and licence from IMDA. The Contractor shall ensure that the frequencies used do not conflict with those of Other contractors. The Contractor shall allow for communication with the S.O. through their radio communication system.
- (c) The Contractor shall provide portable radios for the S.O.. The radios shall be based on current digital technology with open standard protocols meeting DMR (TDMA Division Multiple Access). The digital standards shall last throughout the construction phase and shall be easily upgradable / migrated to any future digital standards when the plant is in initial operations period.
- (d) The portable radios for the S.O. shall be programmable to enable operation on the C1A contractor's frequency as well as the Contractor's frequency. The Contractor shall be responsible for the initial programming of the radios and shall provide the programming software to the S.O. to enable programming to match subsequent contractors' frequencies.

P2.5.11 Buried Services Coordination

- (a) The C1A contractor will manage all Common Areas in the TWRP Area, including all buried services. The Contractor shall comply with the procedures developed by the C1A contractor for buried services including any work nearby or connecting to the buried services in the Common Areas.

P2.5.12 Courier and Mail Delivery Services

- (a) The C1A contractor will provide a centralised location at the NSG Administration Facility for incoming and outgoing mail and deliveries of small packages for the Contractor during the Contract period. The Contractor shall be responsible for the collection of his own mail and packages from the NSG Administration Facility. The Contractor shall comply with the C1A contractor's procedures for administering these services.

P2.6 SITE UTILITIES

P2.6.1 Sanitation

- (a) The C1A contractor will install, operate and maintain a sewage collection system through a combination of gravity and pumped network on the TWRP Area with connection points available to the Contractor (and Other contractors) at designated manholes as shown in the Drawings. Sewage will flow by gravity to storage tanks from where it will be pumped to the PUB sewerage network. The Contractor shall pump the sewage from the Contractor's sewage network to the designated manhole. The Contractor shall note the combined flow from their sewerage network discharged to C1A manholes shall not exceed 781 l/hr. The Contractor shall submit the calculations showing the flow generated from their office area to S.O. for approval. The Contractor shall be responsible for all costs associated with the disposal of sewage for any flows greater than the specified peak flow rate.
- (b) The TWRP Area sewage collection network is planned to be available from the Commencement Date. However, the Contractor shall be responsible for providing his own sewage collection and disposal prior to the availability of the system in the event that the system commissioning is delayed.
- (c) After commissioning, in normal operation, at any time that the TWRP Area sewage system is not available for short periods, the C1A contractor will endeavour to return it to service as quickly as possible. During the period when it is not available, it is the responsibility of the Contractor to manage its own temporary sewage collection and disposal.
- (d) The Contractor shall provide all necessary sanitary facilities on the Site and shall provide all facilities and services necessary to connect to the C1A gravity sewage collection system. The Contractor shall coordinate with the C1A contractor on the connections.

P2.6.2 Potable Water Supplies for the Works

- (a) The C1A contractor will install, operate and maintain a 200 mm diameter potable water reticulation network on the TWRP Area for use by the Contractor and Other contractors.

- (b) The network should be available from the Commencement Date. However, the Contractor shall be responsible for providing its own potable water prior to the availability of the network.
- (c) The Contractor shall connect to the potable water network at the connection point as shown in the Drawings as approved by the S.O.. The Contractor shall provide all facilities and services necessary for the connection. The Contractor shall coordinate with C1A contractor on the connection.
- (d) The C1A contractor will manage the potable water bulk meter account with PUB WSN. The Contractor shall install a sub-meter at the location of connection to the TWRP Area potable water network and arrange with WSN for his own account for supply of potable water to the Site. The Contractor shall allow for at least eight (8) weeks from application to the commencement of potable water supply by WSN. The Contractor shall be responsible to pay all fees, charges and costs for the connection and supply of potable water to the Site. The Contractor shall provide a signed undertaking letter from the C1A contractor to WSN with their application to WSN. The Contractor shall provide to the S.O. periodic estimates of their potable water demand (in m³/day or m³/month) taking into consideration their use of NEWater. Such estimates shall be provided to PUB WSN.
- (e) The Contractor may provide private metered water connections from his supply for his subcontractors working on the Site and shall deal directly with them regarding payments for their water usage.
- (f) In the event that the TWRP Area potable water network is not available, the C1A contractor will endeavour to return it to service as quickly as possible. During the period when it is not available, it is the Contractor's responsibility to provide his own potable water.
- (g) The Contractor shall provide for any temporary potable water required for the testing and commissioning of the Works.

P2.6.3 NEWater Supplies for the Works

- (a) The C1A contractor will install, operate and maintain a 300 mm diameter NEWater reticulation network on the TWRP Area for use by the Contractor and Other contractors for construction activities.
- (b) The Other contractors will provide a 200mm diameter NEWater connection point (subject to NEWater availability and approval by the S.O.) for the Contractor. The NEWater water connection and supply will be available to the Contractor upon handover of the Contractor's Staging area. However, the Contractor shall be responsible for providing his own potable water prior to the availability of the network.
- (c) The Contractor shall connect to the NEWater connection point at the connection location shown in the Drawings as approved by the S.O.. The Contractor shall provide all facilities and services necessary for the connection. The Contractor shall coordinate with Other contractors on the connection.

- (d) The C1A contractor will manage the NEWater bulk meter account with PUB WSN. The Contractor shall install a sub-meter at the location of connection to the TWRP Area NEWater network and arrange with WSN for his own account for supply of NEWater to the Site. The Contractor shall open an account with SP Group for NEWater after signing agreement with PUB WSN for NEWater. The Contractor shall be responsible to pay all fees, charges and costs for the connection and supply of NEWater to the Site. The Contractor shall provide to the S.O. periodic estimates of their NEWater demand (in m³/day or m³/month). Such estimates shall be provided to PUB WSN.
- (e) The Contractor may provide private metered NEWater connections from his supply for his subcontractors working on the Site and shall deal directly with them regarding payments for their water usage. The tariff shall be approved by the S.O..
- (f) In the event that the TWRP Area NEWater network is not available, the C1A contractor will endeavour to return it to service as quickly as possible. During the period when it is not available, it is the Contractor's responsibility to provide his own water.
- (g) The Contractor shall provide for any additional temporary plant water system required for the testing and commissioning of the Works.

P2.6.4 Electricity Supplies for the Works

- (a) The C1A contractor will install, operate and maintain an electrical power network on the site to supply electrical power to the Contractor and Other contractors. The network shall consist of a 22 kV substation, 22/400 V transformers, electrical duct banks, and cables connected to three (3) ring main 400 V substations.
- (b) The electrical power network should be available from the Commencement Date. The Contractor shall be responsible for providing its own electrical power prior to the availability of electric power network.
- (c) The electrical power network for the Contractor's Staging Area will be provided via an electronic type 250A breaker as shown in the Drawings. The Contractor shall be able to draw up to a maximum of 250A, subject to power availability and S.O.'s approval. The Contractor shall provide his own diesel generators for all other site construction activities not supplied by the electrical power network.
- (d) The Contractor shall provide the cables from the 400 V ring main substation to the Contractor's connection point and engage an LEW for their electrical installation. The Contractor's LEW shall submit an application to SP Group for the installation licence, shall apply for the sub-meter for their installation and shall open an account with SP Group. This service shall include cable pulling and installation of sub-meter and termination of cables at the ring main substation connection point. The Contractor shall carry out testing and commissioning of their electrical installations. The Contractor shall coordinate with C1A contractor for the connections.

- (e) The Contractor shall be responsible to pay all fees, charges and costs for the connection and the electricity tariff shall be paid by the Contractor directly to SP Group based on their sub-meter readings.
- (f) The Contractor may provide private metered power connections from his supply for his subcontractors working on the Site and shall deal directly with them regarding payments for their power usage.
- (g) In the event that C1A's electrical power network is not available, C1A contractor will endeavour to return it to service as quickly as possible. During the period when it is not available, it is the Contractor's responsibility to provide their own electrical power.

P2.6.5 Internet Communication

- (a) The C1A contractor will install and maintain a buried internet cable duct bank network on the site consisting of four (4) pipes running from the MDF Room 1 at the base of the Route 2 bridge to three (3) interface panels on-site. The C1A contractor will also provide internet cables from Route 2 bridge to permanent MDF Room 1.
- (b) The Contractor shall have the option to install internet cable service from the MDF Room 1 up to one of the three interface panels to provide internet service to the site office as approved by the S.O.. The Contractor shall be responsible for arranging for the internet cable installation directly with a service provider and for paying all fees, cost and charges for the installation and use of the service. The Contractor will coordinate with the C1A contractor for this installation.
- (c) The permanent MDF Room 1 and internet cable duct network are estimated to be available from Commencement Date. The Contractor shall be responsible for providing its own internet service prior to the commencement of the C1A internet network at his own cost.
- (d) In the event that C1A's internet network is not available, C1A contractor will endeavour to return it to service as quickly as possible. During the period when it is not available, it is the Contractor's responsibility to provide its own internet service.
- (e) The C1A contractor will also install and maintain a wireless Wi-Fi network on TWRP Area to provide wireless service for the Board, S.O. and each contractor's site supervisory staff for the duration of the Contract. The capacity and coverage of the Wi-Fi network will be determined by the C1A contractor following its survey and design of the system; it is expected to be a minimum 10 Mbps for up to 800 users at peak. The C1A contractor, with S.O.'s approval, will establish the scope and procedures for the Contractor and Other contractors to use the system. The C1A contractor, with S.O.'s approval, may impose limits on the number of users of the Wi-Fi system. The service will be available from the Commencement Date.

P2.6.6 Site Fuel Oil Tanks

- (a) The Contractor shall comply with all local regulations applicable to the use and storage of diesel oils, petrol, paraffin and other inflammable fuels and engine driven generators used on the Site. The Contractor shall ensure that adequate precautions are taken against fire. All fuel tanks used on the Site shall be provided with bund walls to contain any oil leaks to prevent any contamination or pollution to the nearby watercourses. Oil interceptors shall also be provided where there is a risk of contamination of the water course arising from accidental spill.

P2.6.7 Compliance with Statutory Regulations

- (a) The Works shall comply in all respects with the relevant Singapore Statutory Regulations, By-Laws and Orders currently in force, and the Contractor's attention is drawn particularly to the requirements of Workplace Safety and Health Act, Power Grid, Environmental Protection and Management Act and Sewerage and Drainage Act etc.
- (b) The Contractor shall be responsible for obtaining the necessary approvals from the relevant departments for all elements of the Works. It shall also be deemed that the rates and prices include all costs to ensure conformity with current regulations of the departments and all other relevant Authorities.
- (c) The supply of all danger notices, labels, warning notices and the like to the requirement of relevant authorities shall be deemed to be included in the rates and prices.

P2.6.8 Working near Electrical Equipment

- (a) Any permanent fencing or other safeguards required to be erected around electrical equipment shall be completed as far as practicable before connection is made to the electricity supply. If this is not practicable, the S.O. may permit the use of temporary fencing or other safeguards.
- (b) If further work adjacent to electrical equipment must be carried out after connection has been made to the electricity supply, the Contractor shall put into operation a "Permit to Work" system to the approval of the S.O..

P2.6.9 Compressed Air

- (a) The Contractor shall supply any necessary compressed air plant and equipment required for erection, testing and commissioning of the Works.
- (b) Diesel engine driven compressors shall not be sited within buildings or in a location that may cause a health hazard to personnel due to exhaust fumes.

P2.6.10 Signboards

- (a) The Contractor shall provide and maintain project signboards including all necessary support, foundation, painting and description of the project to the approval of the S.O.. The signboards shall be dismantled/cleared away from Site as directed within one week of completion of the Contract.
- (b) The Contractor's signboards shall be provided at the TWRP Area, to be placed at the entrance to the Contractor's staging area housing the Contractor's site office.
- (c) The general dimensions and layout of the signboard shall follow the minimum requirements set by the Building and Construction Authority (BCA). The signboard shall bear the name and logo of the project, artworks, the Board, S.O., Contractor and other information as provided by the S.O..
- (d) Drawings, proposed location and calculation of the signboards, signboard foundations and signages are to be submitted to the S.O. for approval before fabrication and installing on Site.
- (e) The signboards shall be dismantled/cleared away whole from Site as directed by the S.O. within one week of completion of the Contract for Part 1.

P2.7 COLLABORATION WITH S.O. AND OTHERS

P2.7.1 Overview

- (a) The Contractor's co-operation with the S.O. and with Other contractors will be essential for the success of the project (Interface work is detailed in Appendix P2B, Interface with Other contractors). The S.O. will convene regular and ad hoc meetings, in which the Contractor shall participate during the Contract period.

P2.7.2 Attendance by the Board's Representative at Meetings

- (a) The Board reserves the right to attend any or all meetings between the S.O. and the Contractor, Other contractors, subcontractors and suppliers.
- (b) The Board reserves the right to convene additional meetings at his own instigation with the Contractor and S.O..

P2.8 PUBLIC RELATIONS

P2.8.1 Overview

- (a) TWRP is a highly visible project in Singapore and around the world. Much interest in the project is anticipated from the local community and global water industry professionals. In addition, occasional VIP visits are to be expected. The Contractor shall work with the S.O. and the Board to provide the following events/services.

P2.8.2 Official Visits

- (a) The Contractor shall note that from time to time, official visits may be conducted on Site to brief visitors about the project. The Contractor shall give due assistance which may be required of him for such official visits.

P2.8.3 Press and Public Relations Matters

- (a) The Contractor shall designate a senior member of his Contract team to act as his Public Relations (PR) Officer to liaise with the S.O. and DTSS2 Public Relations Team on all press and public relations matters during the Contract period.
- (b) All press releases, press statements, articles or printed material prepared by the Contractor shall be cleared with the Board and S.O. prior to publication or release to the news media.
- (c) All press queries relating to the Contract received by the Contractor shall be referred to the Board and S.O. for clearance. The Contractor is not allowed to be interviewed by the press or divulge any information to reporters without first seeking clearance from the Board and S.O..
- (d) The use of the Board or S.O.'s logo in the Contractor's publication (if any) shall be subject to the written approval of the Board or the S.O..
- (e) The Contractor shall extend to the S.O. all the necessary assistance and co-operation regarding requests for photo-taking, video-taking and visits to the construction site by the Board's and/or S.O.'s official photographer(s) or appointed film-maker(s).
- (f) The Contractor is required to include a section on community relation matters in its monthly report to the S.O..
- (g) All public complaints shall be thoroughly investigated and acted upon by the Contractor on an urgent basis. The S.O. will be informed whenever any complaint is received and advised on the proposed action taken.
- (h) The Contractor shall provide support and make available appropriate materials and information for the Board's public engagement use throughout the duration of the Contract period. The Board's public engagement will include events, presentations, pamphlets, posters, brochures, signs, newspaper articles, photographs, site visits, etc..

P2.8.4 Advertisements

- (a) The Contractor shall not, except with the written authority of the S.O. exhibit or permit to be exhibited on the Site any advertisement board. Any such advertisement may also be subject to the approval of the local planning authority before it is put up and it shall be removed if the S.O. so demands.

P2.9 STORM WATER DRAINAGE AND EARTH CONTROL MEASURES**P2.9.1 General**

- (a) This section describes the minimum requirements for storm water drainage and earth control measures (ECM).
- (b) The work to be carried out under the Contract shall comprise of the following:
 - (i) Provision of elements of the permanent drainage system. The Contractor shall construct, maintain and monitor the permanent drainage system during the Contract period;
 - (ii) Provision of temporary drainage systems to manage storm water runoff for the Site during the Contract period. The Contractor shall design, construct, maintain and monitor the temporary drainage systems during the Contract period; and
 - (iii) Provision of ECM on Site during the Contract period. The Contractor shall engage a Qualified Erosion Control Professional (QECF) to design the ECM on Site. The Contractor shall implement, maintain and monitor the approved ECM Plan on Site.
- (c) All permanent drains shall be constructed based on a 1 in 600 slopes unless otherwise stated or approved by the S.O..
- (d) The Contractor shall construct the storm water drains as per the BCA approved drawings, which shall be provided to the Contractor by the S.O..
- (e) Should there be any obstructions or site constraints to the construction of the permanent drains, the Contractor shall submit a proposal with the change to the design to the S.O. for review and approval. The proposal should include hydraulic calculations to show that the hydraulic capacity of the proposed changes is greater or equal than the original design. The proposal should include hydraulic modelling results if deemed necessary.

P2.9.2 Temporary Drainage

- (a) The Contractor's work areas shall be shared with the C4A contractor and Other contractors.
- (b) Although the above areas are shared areas, where necessary, the Contractor shall design, construct and maintain for the duration of the Contract period the following:
 - (i) The temporary drains within the Contractor's staging areas and work areas in this Contract shall be deemed to be priced under the Contractor's preliminaries.

- (ii) The Contractor shall divert, demolish and reconstruct the temporary drains within the Contractor's staging areas and work areas in this Contract as necessary during the Contract period.
 - (iii) The Contractor shall be responsible to ensure that all the storm water runoff generated from the Site is discharged to the approved discharge locations within the regulatory requirements.
- (c) The Contractor shall design the temporary drainage system to the Code of Practice on Surface Water Drainage.
- (d) The slope of the temporary drains shall not be gentler than a 1 in 1000 slope.
- (e) The Contractor shall collect and treat 100% of the storm water runoff generated within their Site Boundary up to a 5-year return period storm event. This is even if the storm water is generated from paved surfaces and meets discharge limits. The storm water runoff collected would need to be treated and discharged up to 10 hours in anticipation of the next storm event.
- (f) The Contractor shall develop a temporary Site Drainage Plan at every stage of the Works, which will need to be approved by the S.O. prior to implementation on Site. The plan will need to be constantly updated to suit the multiple stages of work throughout the Contract period. The Contractor shall seek approval from the S.O. and the C1A contractor on the location and peak flows expected at each discharge points outside the C4B Site Boundary.
- (g) The Contractor shall have no more than (three) 3 discharge points to the C1A contractors' drainage network. Direct discharge into the sea is not permitted.
- (h) The Contractor shall seek clarification from the S.O. on any assumptions he intends to make or any further information he requires with regards to the drainage design. The Contractor shall submit the Site Drainage which would contain the temporary drains design calculations for approval by the S.O. prior to implementation on site.
- (i) The Contractor shall operate and maintain the temporary drainage systems within the Site.
- (j) The Contractor shall keep all temporary drains clean and in good condition at all times. Drain cleansing shall be carried out at least monthly to remove any silt built up in the drains. Any drain blockage shall be cleared and cleaned by the Contractor within 24 hours or as soon as practical after the blockage is observed.
- (k) Sedimentation ponds / tanks shall be inspected after each major storm event or on a weekly basis to ensure that inlet and outlet pipes are not obstructed. Every six (6) months, ponds shall be drained, inspected, cleaned and repaired as necessary to re-establish the side slopes, bottom grade, and lining systems.

- (l) Damage to the drains within the Site shall be immediately repaired after the damage is identified. The repair shall be carried out to the satisfaction of the S.O.. The repair shall be completed within 3 working days. Temporary measures to ensure the functionality of the drainage system shall be put in place if necessary. This may include the diversion of the drains or mobilisation of storm water pumps.
- (m) The Contractor shall be responsible for the demolition of the temporary drainage system within the Contractor's staging areas and work areas in this Contract.

P2.9.3 Earth Control Measures

- (a) The Contractor shall implement necessary earth control measures within his Site. For shared staging and work areas with Other contractors, where necessary, the Contractor shall implement necessary earth control measures.
- (b) The ECM implemented on Site shall comply with the requirements as specified in G9 Environmental Considerations and Protection and shall include the following as the minimum:
 - (i) Perimeter cut off drains to be provided;
 - (ii) Silt fences and lined silt trapping devices along the cut off drains;
 - (iii) Erosion control blankets for bare earth surfaces;
 - (iv) Sedimentation basins / tanks for the collection and treatment of silty waters prior to discharge to the approved discharge locations;
 - (v) Silty water treatment units for the treatment of silty waters prior to discharge to the approved discharge locations; and
 - (vi) Proposed monitoring locations with CCTVs and online total suspended solids (TSS) meters for the discharge of storm water at the approved discharge locations.
- (c) The Contractor shall incorporate the ECM elements into the Site Drainage and ECM Plan.
- (d) The Contractor shall allow in his Contract Sum for the total management of all ECM provisions to manage the erosion and sediment control at Site over the Contract period.
- (e) Integrated CCTV and Online TSS Monitoring System.
 - (i) The Contractor is required to implement and maintain an Integrated CCTV and online TSS monitoring system.
 - (ii) The CCTV monitoring units installed at the discharge locations shall be weatherproof and able to communicate wirelessly to the integrated CCTV and online TSS monitoring system.

- (iii) The system shall be a web-based application that displays the live feeds and real-time readings of the CCTVs and online TSS meters of each individual unit on Site. The System should be easily accessible on computers and mobile devices without the need to install additional software.
- (iv) The interface shall be graphical and clearly showing the current location of each individual monitoring unit on Site. As the monitoring unit's locations are expected to change on Site as work progresses, the location map on the graphical interface shall be updated accordingly within 5 days of any change.
- (v) The system shall be set up to store the CCTV snapshots and TSS readings at 5 minutes intervals over a 30-day period.
- (vi) The system shall be capable of exporting daily, weekly and monthly graphical reports. These reports are to be submitted to the S.O. or the Board within 2 days when requested.
- (vii) The CCTV units are required to be operational at night and at locations where there is low light. Sufficient lighting is to be provided to ensure that the CCTV footage is usable. CCTV requirements should follow the CCTV requirements listed on <https://www.pub.gov.sg/Documents/CCTV%20requirements.pdf> at the minimum.
- (viii) The system is to be set up to send out short message service (SMS) alerts to users if TSS readings of any unit exceeds the allowable limits. The system should also be programmed to send out SMS alerts should there be any fault in any of the individual units.
- (ix) The Contractor shall alert the S.O. of any non-compliance of the discharge TSS concentration or malfunction in any equipment within one (1) hour. The Contractor is to investigate the non-compliance and ensure that the appropriate rectification measures are implemented within (twenty-four) 24 hours.
- (x) The individual online TSS monitoring units are to be calibrated monthly. The Contractor is to submit the calibration reports to the S.O. for record.
- (xi) The Contractor shall provide the necessary standby and maintenance requirements such that the Integrated CCTV and online TSS monitoring system is not down for more than 24 hours.

P2.9.4 Site Drainage and ECM Plan

- (a) The Contractor is to develop a Site Drainage and ECM Plan for every stage of his works. The plan shall take into consideration:
 - (i) The phasing of construction works, especially for excavation works;
 - (ii) Changes in ground levels;

- (iii) Changes to road alignments and levels; and
 - (iv) Approved discharge points.
- (b) The Site Drainage and ECM Plan shall include:
 - (i) Current work areas;
 - (ii) Drainage channels provided (including invert levels, drain slopes, drain widths and drain cope or soffit levels);
 - (iii) Delineated sub-catchment areas contributing to each segment of the drainage network (including areas and dimensions of the catchments);
 - (iv) The runoff coefficient values for each of the sub-catchments;
 - (v) Drainage direction of flow;
 - (vi) ECM sedimentation basins (including effective volume of basin, basin dimensions);
 - (vii) ECM silty water treatment units (including treatment capacities in m³/s and discharge points); and
 - (viii) ECM integrated CCTV and TSS monitoring locations.
- (c) The Site Drainage and ECM Plan shall be appropriately and clearly labelled so that each individual element can be clearly identified and called out.
- (d) The corresponding hydraulic calculations and hydraulic modelling results (if necessary) supplementing the Site Drainage and ECM Plan shall be submitted together.
- (e) The corresponding ECM Design Calculations, Erosion Control Plan, Sediment Control Plan and Site ECM Management System as specified in General Specification G9 Environmental Considerations and Protection shall be submitted to supplement the Site Drainage and ECM Plan.
- (f) The corresponding maintenance and monitoring regime supplementing the Site Drainage and ECM Plan shall be submitted together.
- (g) The Site Drainage and ECM Plan would need to be approved by the S.O. and any other relevant authorities (i.e. PUB, NEA, MPA, SP Group, TPG) prior to implementation on Site. The Contractor shall factor in sufficient lead time for the planning, rework and approval process.
- (h) Within three (3) weeks of the Commencement Date, the Contractor shall submit the preliminary Site Drainage and ECM Plans to the S.O. for review.
- (i) The Site Drainage and ECM Plan shall be endorsed by the Contractor's PE and QECF.

P2.10 CONTRACTOR'S ORGANISATION

P2.10.1 Contractor's Organisation

- (a) The Contractor shall submit a diagram showing the organisational structure for the management of the contract, with locations, names and position titles of staff and their line and staff relationship. The diagram shall include associate organisations and subcontractors and show clearly the individuals and lines of responsibilities linking the various groups. It shall also indicate the persons designated as contacts with the S.O.. If staff are not manning the positions for the entire duration of the Contract, these shall be explained in the narrative, indicating in full detail when they are to be mobilised and demobilised. Staff identified as key staff in compliance with the Tender requirements including the Evaluation Criteria shall be included and clearly identified:
 - (i) The names, qualifications, positions, current resumes and relevant certifications of key executives, supervisory and engineering staff to be employed full time on the project shall be included with the submission including principals of subcontractors; and
 - (ii) A narrative describing the sequence, nature and inter – relationship of the various main Contract activities including timing and exchange of information.
- (b) The S.O. will review the Contractor's proposed organisation and shall have the right to require the Contractor to make amendments as deemed necessary by the S.O..
- (c) The Contractor shall provide a competent Project Management Team to be based at the Site, from the commencement until the completion of the Works, with the necessary expertise and experience to represent the Contractor, liaise with and receive instruction from the S.O. pertaining to the Contract.
- (d) The Contractor shall engage qualified and competent professionals who have the relevant experience to carry out all the Works in accordance with the Specifications.
- (e) Any deviations in Qualification and Experience for key personnel specified in Volume 1 Annex I shall be subject to S.O.'s discretion and approval.
- (f) Except as the Board may otherwise agree, no change shall be made in the personnel proposed by the Contractor and approved by the S.O.. If, for any reason beyond the reasonable control of the Contractor, it becomes necessary to replace any of the personnel, the Contractor shall forthwith propose justification and provide an appropriate replacement acceptable to the Board.
- (g) If any person specified in the Tender submitted by the Contractor is found by the S.O. to be incompetent in discharging his assigned responsibilities or is a security and / or safety risk or has misconducted himself, the S.O. may request the Contractor to forthwith provide a replacement with such qualifications and experience as may be acceptable to the Board.

- (h) Except as the Board may otherwise agree, the Contractor shall bear all costs arising out of or incidental to any replacement of personnel.
- (i) The Contractor shall keep accurate and systematic accounts and records in respect of the services, in such form and detail as mutually agreed by the parties.
- (j) The Contractor shall provide a suitably qualified and experienced Project Manager whose appointment shall be subject to the approval of the S.O.. The Project Manager shall be full time on Site during the Contract period and devote himself full time to the Project, commencing no later than fourteen (14) calendar days from the Commencement Date and finishing only after the S.O. has certified the Works to be fully completed. The Contractor shall also provide intermittent attendance of the Project Manager as and when required by the S.O. during the Defects Liability Period.
- (k) All correspondence and communication between the S.O. and the Contractor will be directed through the Project Manager.
- (l) The Contractor shall provide a Licensed Electrical Worker (LEW) who shall supervise the electrical design testing and commissioning works. This shall include cable diversion, cable / excavation works next to existing “live” cables. The LEW shall be responsible to certify fitness for turn on of the electrical installation and handover of the completed Works. In addition the LEW shall be required as and when to carry out all switching operations to facilitate testing and commissioning and any other related construction activities related to the Works.
- (m) Designs, drawings and documents that are required to be submitted to the S.O. for approval shall be considered as bona fide by the S.O. only if they have been submitted by the Project Manager. Prior to submission to the S.O., in addition to endorsement of a PE Singapore, the Project Manager shall have checked the design, drawing or document, whether initiated by the Contractor himself or by one of his subcontractors, and the design, drawing or document shall be clearly identified as to its location in, or application to the Works.

P2.10.2 Minimum Project Personnel

- (a) As a minimum, the following project personnel shall be provided for the duration of the Contract period and Contractor shall include in the rates and prices.
- (b) Unless stated otherwise, the following personnel listed below shall be dedicated **FULL TIME** to the project for the duration of the Works. Attendance at Site shall be fully logged in the electronic security systems.
- (c) A different person shall be nominated for EACH position. One person cannot fulfil multiple positions.
- (d) Minimum Qualification and Experience required for key personnel are provided in Volume 1 Annex I.

Table P2-2: Minimum Project Personnel

Position	Mobilisation
Project Director	Minimum 4 hours a week on Site
Project Manager	FULL TIME stationed on Site
Project Controls Manager	FULL TIME stationed on Site
Construction Manager	Be available during planning and be FULL TIME stationed on Site during construction
Earth Control Measures Officer (ECMO) /Environmental Control Officer (ECO)	FULL TIME stationed on Site during construction
Quality Assurance/Quality Control Manager (QAQCM)	FULL TIME stationed on Site during construction
Building Information Management System Manager (BIMM)	FULL TIME during design and construction
Building Information Management Authoring Leads	FULL TIME during design and construction
Design Manager	FULL TIME during design and to be on Site when required during construction and Testing and Commissioning Period
Civil Engineering Manager	FULL TIME during the design and FULL TIME stationed on Site during the construction.
Bored Piling Engineer (BPE)	FULL TIME during the execution of the bored piling works
Electrical Engineering Manager (EEM)	To be on Site for all electrical work
Licensed Electrical Worker (LEW)	To be on Site for all power turn on activities including connections of Other contractors to the power supply
Mechanical Engineering Manager (MEM)	FULL TIME during the design and FULL TIME stationed on Site during the installation and testing of the mechanical works.

Position	Mobilisation
Control System Engineering Manager (CSEM)	FULL TIME stationed on Site during the software development, FAT, SAT and Testing and Commissioning Period
Senior Planner	FULL TIME stationed on Site
Risk Manager	Minimum to be on Site for risk facilitation and risk management activities.
Interface Manager	Be available during planning and be FULL TIME stationed on Site during construction
DfS Professional	Minimum to be on Site for risk facilitation and risk management activities.
Document Controller	FULL TIME stationed on Site
HSE Manager	Refer to General Specifications G2 and Particular Specification P9; be available on Site as required by the S.O. during design and planning and be FULL TIME stationed on Site during construction
Registered WSH Officer (WSHO)	Refer to General Specifications G2 and Particular Specification P9; be available on Site as required by the S.O. during design and planning and be FULL TIME stationed on Site during construction
Registered WSH Coordinator (WSHC)	Refer to General Specifications G2 and Particular Specification P9; be FULL TIME on Site during construction
Testing and Commissioning Manager	Minimum to be on Site FULL TIME for Testing and Commissioning Period
Testing and Commissioning Specialists (THP)	FULL TIME stationed on Site for Test & Commissioning Period
Testing and Commissioning Specialists (SSN)	FULL TIME stationed on Site for Test & Commissioning Period
Operations and Maintenance Manager	FULL TIME stationed on Site for the Initial Operations and Operations and Maintenance Periods

Position	Mobilisation
Supervisors and Technicians for the Operations and Maintenance Team – Refer to Particular Specification P12 Operations and Maintenance	FULL TIME stationed on Site for the Initial Operations and Operations and Maintenance Periods

- (e) The above specified list of personnel shall be a minimum and shall not be limiting. The Contractor shall propose other construction professionals with relevant qualifications for the proper execution, supervision and management of the Works as required.
- (f) In addition, the Contractor shall provide the following personnel who shall be on Site where required to support the construction works:
 - (i) Quantity Surveyors; and
 - (ii) Registered Surveyor.
- (g) In addition to the HSE personnel listed above, the Contractor shall provide the following full time competent HSE personnel for implementing effective HSE programs and risk management at the worksite. This Specification shall be read in conjunction with General Specification G2.23 Health, Safety and Environment:
 - (i) WSHC per shift (FULL TIME on Site during construction);
 - (ii) WSHS (FULL TIME on Site during construction); and
 - (iii) HSE Administrative Assistant (FULL TIME on Site during construction).
- (h) The Contractor shall inform the S.O. prior to any replacement of personnel due to termination, resignation, etc. Replacement personnel shall be subject to the approval of the S.O.. The Contractor shall obtain approval from the S.O. for his site personnel who is on leave or attending courses for a period of 5 or more days. The S.O. reserves the right to request the Contractor to remove and replace any staff if he does not perform to the satisfaction of the S.O..
- (i) The Contractor shall note that if the S.O. deems that the level of supervision is inadequate or when certain portion of the Works require more supervision effort, the Contractor shall strengthen his project team at his own cost.
- (j) In addition to the measures that may be taken by the authorities against the Contractor, the Board is entitled to impose an administrative charge on the Contractor in a specified amount for the duration the Contractor fails to provide and maintain the following manpower resource in accordance with the requirements of the accepted organisation chart, the relevant applicable laws, and/or the Contract, whichever is more stringent:

Table P2-3: Administrative Charges

Required Manpower Resource	Administrative Charge
Project Manager	S\$15,000 / week
Construction Manager	S\$13,000 / week
Testing and Commissioning Manager	S\$13,000 / week
Building Information Management System Manager (BIMM)	S\$15,000 / week

P2.10.3 Engagement of Qualified Person(s) for Design

- (a) The Contractor shall engage Professional Engineer(s) as required to serve as qualified persons (QP) for the design of any elements of the Permanent and Temporary Works detail designed by the Contractor, which shall comply with the requirements of the Building Control Act, the Building Control Regulations and all the relevant Authorities.
- (b) The Contractor shall prepare and endorse the necessary calculations, details and drawings and submit it to the S.O. who will pass it to the Board's AC for checking. The Contractor shall address any design comments raised by the AC. Once endorsed by the AC, the Contractor shall submit the design to BCA and obtain the approval for construction.
- (c) Upon getting BCA Approval, the Contractor is required to submit a copy of approved drawings and calculations to the S.O..

P2.10.4 Contractor Supervision of Construction of ERSS

- (a) The QP Supervision engaged by the Contractor for the ERSS shall be responsible for the proper construction of all ERSS Works in accordance with the approved plans. The Contractor's QP Supervision team shall include for QP(S), QP(S)-Geo, RE and RTOs. The QP shall check the instrumentation and monitoring readings regularly and submit the report to the S.O. according to the regulations.
- (b) The Contractor's QP Supervision shall undertake the supervision and certification of all ERSS work including the deployment of his own site supervision staff to ensure compliance with the Contractor's QP Design requirements under his design and build obligations.
- (c) The Contractor shall extend full assistance and co-operation to the S.O. and his representatives at all times. All works shall be carried out only with the concurrence of and to the satisfaction of the S.O..

- (d) The S.O. will closely monitor the progress of the construction of the ERSS. The presence of the S.O. and his representatives during the construction of the ERSS shall not absolve the Contractor of his responsibility to ensure proper execution and supervision of the Works.

P2.10.5 Engagement of QECP and ECMO

- (a) The Contractor shall engage a qualified QECP to design and make submissions to the Board's Catchment and Waterways department to obtain approval for the proposed earth control measures (ECM) for the Site. The Contractor shall ensure that the QECP makes expeditious submissions to approving authorities to ensure no delay to the project execution.
- (b) The ECM plan for the Site shall include any areas for stockpiling of the spoil for future backfilling work. The Contractor shall allow for the use of Erosion Control Blankets (ECB) to minimise erosion of earth on the slope embankment of the open excavation and on the slopes of the stockpile.
- (c) The Contractor's QECP shall be responsible for the design of the Site wide drainage system for the interception, control and management of surface runoff, collection and discharge of water from direct rainfall.
- (d) The Contractor's QECP shall allow for relocation of the sump pits, cut off drains, pumps and pipe networks, sedimentation ponds and all associated works in response to the changing needs of the ERSS system, i.e. when carrying out stage excavation down to formation level and when carrying out stage backfilling up to the ground level. The Contractor's QECP must provide the necessary revisions to the ECM plan and the drainage plan and obtain the necessary approvals from the authorities and the S.O. for every significant change to the drainage scheme.
- (e) The Contractor shall appoint or engage a qualified ECM Officer (ECMO) for the project. The ECMO shall implement the ECM measures according to QECP plan, devise ECM operations, set up maintenance and inspection programme, and assist the QECP to monitor ECM effectiveness throughout the various stages of work.

P2.11 CONTRACT ADMINISTRATION

P2.11.1 Reporting Progress

- (a) To provide a qualitative and quantitative record of work progress against plan, contract changes, forecast and monthly progress payment related items; the Contractor is required to submit a suite of documents at the end of each reporting month, referred to as the Monthly Progress Report as specified in the General Specification G2.6 Progress Reporting.

P2.11.2 Recording Construction Progress

- (a) Progress Digital Media
 - (i) The Contractor shall provide progress photographs. One set of progress photographs shall be submitted weekly.
 - (ii) The Contractor to provide for all necessary poles for mounting equipment, supports, and foundations as required.
 - (iii) The Contractor shall provide the time lapse videos on a monthly basis showing the monthly progress and the progress from the Contract Commencement Date.

P2.11.3 Method Statements

- (a) The Contractor shall prepare and submit to the S.O. detailed method statements for all aspects of his work and especially for aspects of his work which may affect the work of Other contractors on the Site. The method statements shall be submitted for approval by the S.O. at least 28 days before the work covered by the method statement is programmed to commence. Aspects of work for which method statements shall be prepared are as follows:
 - (i) Work which may affect access to the Works;
 - (ii) Work which may affect the electricity supply system (both the temporary and permanent power supply);
 - (iii) Work which may influence the entrance guardhouses, Route 2 and Route 3;
 - (iv) Work which involves filling any part of the works with water or passing water through any part of the Works as part of the testing and commissioning;
 - (v) Each method statement shall comprise a step-by-step schedule of specific operations and activities with descriptions, dates, times and durations for each step. It shall highlight specific points in the programme and include contingency plans for emergency reinstatement of plant including "point of no return" and "earliest start time" of modified or new plant. The statement shall clearly state who and what organisation will carry out work and precisely when and what support or provisions are to be made by others outside the Contract. The statement shall be supported by sketches, diagrams or other supportive detail to enable a clear understanding of the method and significance of each step of the operation or work; and
 - (vi) No work which affects the work of Other contractors shall be carried out until a method statement has been submitted to and approved by the S.O..
- (b) The Contractor shall refer to the General Specification G2.23 Health Safety and Environment which refers to method statements required for safety critical activities.

P2.11.4 Forms for Monthly Statements

- (a) Monthly statements for progress payments to the Contractor submitted under the Conditions of Contract shall be presented on forms to the requirements of the S.O.. Such forms shall be provided by the Contractor and shall be printed to the approval of the S.O. as soon as practicable after the acceptance of the Tender.

P2.11.5 Standard Forms

- (a) The Contractor shall submit the templates for the standard forms for the Works in compliance with the requirements specified in General Specification G1.2 Information Management.
- (b) The Contractor shall establish a document control and monitoring system for the following standard forms used to control queries and requests:
 - (i) RFIW – Request for inspection of works;
 - (ii) RFI – Request for information / clarification;
 - (iii) RAM – Request for approval of materials; and
 - (iv) TPW – Temporary permit to work.
- (c) The naming and number convention for each set of forms shall comply with General Specification G1.2 Information Management.
- (d) The status report of submission and responses shall be provided to the S.O. on a weekly basis via the TWRP PMIS.

P2.12 PROGRAMME FOR THE WORKS

P2.12.1 Overview

- (a) The Contractor shall comply with Clause 9 of the PSSCOC standard conditions of contract and General Specification G2.2 Project Controls and Programme for the Works when preparing and submitting the Programme for the Works.
- (b) The monthly updating of the Programme for the Works is necessary to verify the estimate or invoice upon which progress payments will be made. If the Contractor fails to provide sufficient information for the S.O. to approve a progress update or a revision to the programme for the Works, the Contractor will not be entitled to progress payment until the necessary information is furnished and the S.O.'s approval is obtained.
- (c) The Contractor shall develop the Programme for the Works in accordance with the breakdown structure outlined in Appendix P2A.

- (d) The Contractor shall provide the full list of activities duration calculation of the Programme for the Works with its assumed quantities and assumed productivity as basis. This spreadsheet shall be submitted together with Basis of the Programme for the Works outlined in G2.2.7.
- (e) The Programme for the Works shall be updated to incorporate System and Sub-System completion activities from installation to testing and commissioning. This is to be completed by March 2024 (date provisional only and to be confirmed by the S.O. after Contract award).

P2.12.2 Key Project Milestones

- (a) The Contractor shall achieve the key project milestones for the Works as stated in table below. The Contractor shall sequence their construction works and provide adequate resources to achieve these key project milestones.

Table P2-4: Key Project Milestones

Milestone No.	Key Project Milestones	Days from Commencement Date
1	Transfer of work area from C4A to the Contractor - THP Courtyard	300
2	Enable access to work area from C4A to the Contractor - Biosolids Building Rooms	300
3	Compressed Air System to provide air supply to C4A	660
4	*Successful completion of Pre-Commissioning Test for Digestion Pre-treatment Facility, Sidestream Nitrogen Removal Facility and Pipe Rack together with Pipes	852
5	*Completion of the whole of the Works (Include Plant Performance Test)	1,181

Notes:

Milestone marked * is a LD Milestone (refer to Volume 1 Appendix to the Conditions of Contract).

- (b) The Contractor shall note that the definition of the term 'transfer of work area' as follows:
 - (i) The civil and structural element of the works (C&S) of the specified work area to be substantially completed by C4A contractor to the satisfaction of the S.O.; and

- (ii) The successful completion of all testing and commissioning requirements as specified in G2.26 for Stage 1 and 2 Testing for the C&S specified work area by C4A contractor.
- (c) The Contractor shall note that Staging Area as shown in the Drawings will be available to the Contractor to take over, in full or partial, for from Dec 2022 onwards. The Contractor shall note that the date is provisional and indicative, it will be confirmed after Contract award by the S.O..
- (d) The following shall occur for Milestone No. 1:
 - (i) The S.O. will facilitate the handover of the specified work area from the Other contractors to the Contractor. The Other contractors shall coordinate with the S.O. and the Contractor to ensure the handover of the work area including access to the work area. The Other contractors are to document the handover of the specified work area. The document should minimally indicate when the area is handed over, the extent of the area handed over and endorsement/acknowledgement by both parties with date specified. Once handed over, the work area will no longer be under the Other contractors' charge.
 - (ii) The Contractor shall note that the specified work area remains as shared work areas after the transfer of work area. The Other contractors will co-ordinate and work with the Contractor and the S.O. for access to the shared work areas for the successful completion of their remaining TWRP Works.
 - (iii) The Contractor shall note that these transfer dates are provisional and subject to agreement between the Contractor and Other contractors and approval by the S.O..
- (e) Milestone No. 2 is a provisional date when the Contractor can commence work in various Biosolids Building Rooms. These rooms will remain as shared work areas and will be under the Other contractors' charge. The Contractor shall coordinate and work with Other contractors and the S.O. for access to the shared work areas for the successful completion of the related Works.
- (f) Milestone No. 3 includes the supply of instrument air to C4A for their works. Refer to Particular Specification P4 Process – Mechanical Works for Biosolids Building Compressed Air Requirement.
- (g) Milestone No. 4 includes the following:
 - (i) Completion of Pre-commissioning Tests (Stage 2 as specified in General Specification G2.26) for Digestion Pre-treatment as stipulated in the Contract and to the S.O. satisfaction to receive sludge for commissioning. The Contractor shall note that sludge may not be available prior to this date (subject to confirmation of the S.O.);

- (ii) Completion of Pre-commissioning Tests (Stage 2 as specified in General Specification G2.26) for Sidestream Nitrogen Removal Facility as stipulated in the Contract and to the S.O. satisfaction to receive centrate for commissioning. The Contractor shall note that centrate may not be available prior to this date (subject to confirmation of the S.O.); and
- (iii) Completion of Pre-commissioning Tests (Stage 2 as specified in General Specification G2.26) for the WSW41 Sitewide Piperack and the pipework on the piperack between Biosolids area and ILM area.
- (h) Certificate of Substantial Completion for the Whole of the Works will be issued to the Contractor upon substantial completion of Milestone No. 5 (the Whole of the Works) as stipulated in the Contract and to the S.O.'s satisfaction. Milestone No. 5 includes the handover of staging area, which the S.O. will facilitate, from the Contractor to C6B contractor.
- (i) Steam is supplied from IWMF at the tie-in location. The Contractor shall coordinate with the S.O. for steam supply to the Digestion Pre-treatment facility after Contract award by the S.O..
- (j) The Contractor shall provide to the S.O., in a timely manner, prior to the completion of the Works, all necessary information related to the Works required to enable the S.O. to apply for and obtain BCA TOP (including addressing the Registered Inspectors comments). The Contractor shall be required to co-ordinate with the S.O. and Other contractors to achieve TOP. The Contractor shall note that the cost of providing such information and co-ordination shall be deemed to be included in the Contract sum.

P2.13 PROJECT CONTROLS

P2.13.1 Overview

- (a) The Board and the S.O. promote the use of consistent program management and control methodologies across all contracts for the successful delivery of the TWRP Project. Successful project delivery depends on consistent and proven program controls processes, structure, and tools to disseminate and integrate timely, accurate progress and performance information. Refer to General Specification G2.2 Project Controls and Programme to the Works.

P2.13.2 Risk Management

- (a) The Contractor shall refer to the preliminary risk registers developed by the S.O. in Volume 6, which is provided for information and shall not be relied upon as a true reflection of Contract risk. The Contractor shall consider further the Contract risk and develop a comprehensive Risk register with identified mitigation measures for the risks identified. The Contractor shall add risks or amend the preliminary risks as he sees fit. The Contractor shall own the Risk register and shall maintain and update it as the Contract progresses. The Risk register will be discussed regularly between the Board, the S.O. and the Contractor.

- (b) The Contractor's risk management processes shall comply with the requirements set out in General Specification G2.2 Project Controls and Programme to the Works.

P2.13.3 Cost Management and Expenditure Forecasting

- (a) The Contractor shall submit to the S.O. within four (4) weeks of acceptance of the Programme for the Works, an expenditure forecast for each reporting period of the Contract through to completion of the whole of the Works and any OM period. The expenditure forecast shall be a comprehensive estimate of the timing and quantum Board's payments to the Contractor based on information available at the time of estimation. The expenditure forecast shall be updated by the Contractor from time to time when required by the S.O..
- (b) A breakdown structure shall be developed by the Contractor in line with the Schedule of Prices (SOP) to report project expenditure. The Contractor shall develop a structure in consultation with and subject to acceptance by the S.O., to ensure compatibility with the overall project-wide breakdown structure and BIM reporting requirements. The Contractor shall comply with the expenditure management requirements outlined in the General Specification G2.2 Project Controls and Programme to the Works including requirements to comply with linking the BIM model (5D) to show cost breakdown.
- (c) The Contractor shall comply with the cost reporting requirements outlined in the General Specification G2.6 Reporting and Progress Payment.

P2.13.4 Project Controls Meetings

- (a) A calendar of Project controls and Project management meetings related to each contract will be provided to the Contractor by the S.O. upon Contract award. The document will include frequency, attendance and reporting requirements. As a minimum the Contractor to allow for the following:
 - (i) Monthly Progress meeting attended by the Contractor's senior management team in the TWRP project;
 - (ii) Bi-weekly Project Controls meetings;
 - (iii) Monthly Contract administration meetings to review and evaluate progress claims;
 - (iv) Meetings outlined in the General Specification G2.2 Project Controls and Programme for the Works; and
 - (v) Other routine meetings deemed necessary by the S.O..

P2.14 DESIGN BY CONTRACTOR**P2.14.1 Contractor's Responsibility for Developing Design**

- (a) Responsibility for Detailed Design of Works
- (i) The Contractor is responsible for developing the reference design for construction, subject always to the approval of the S.O.. The required performance and workmanship of all facilities and equipment is described in the Specifications. The Contractor's detailed design shall comply with the Specifications.
 - (ii) The P&IDs show the scope of the Works including installations outside the scope of works for C4B. The Contractor is to review the Specifications together with the Drawings to understand the Contractor's scope, in relation to the Other contractors and the interface points. The Contractor shall note the Appendix P2B Interface with Other contractors and the Tie-in points (TP) shown in the Drawings.
 - (iii) The P&IDs show the reference design with the minimum process requirements and functionality. The Contractor shall develop the P&IDs and any additional pipework, valves, instrumentation and associated equipment required, to provide a complete system for the Works, shall be included for in the Contract Sum.
 - (iv) The Contractor shall conduct HAZOP and ALMOP (Access Lifting Maintainability and Operability) workshop(s) for the Contractors scope of works at his own cost. This shall include the provision of a facilitator for conducting the HAZOP and ALMOP workshops subject to the approval of the S.O.. The HAZOP and ALMOP workshop(s) shall include the S.O. representatives. The Contractor shall provide the HAZOP and ALMOP report(s) to the S.O. for approval. Any agreed additional modifications to the process required as a result of the HAZOP and ALMOP report(s), shall be at the Contractor's own cost.
 - (v) If the Contractor modifies the P&IDs after the HAZOP and ALMOP workshops, the Contractor shall submit these modification(s) to the S.O. for approval. The S.O. will decide if the modification is substantial and for all substantial modifications, the Contractor shall conduct additional HAZOP and ALMOP (Access Lifting Maintainability and Operability) workshop(s) at his own cost. The HAZOP and ALMOP workshop(s) shall include the S.O. representatives. The Contractor shall provide the HAZOP and ALMOP report(s) to the S.O. for approval. Any agreed additional modifications to the process required as a result of the HAZOP and ALMOP report(s), shall be at the Contractor's own cost.

- (vi) The Drawings including the Electrical Single Line Diagrams are based on the reference design presented herein at the time of tender, which describe the minimum requirements to be provided by the Contractor for his scope of work. The Contractor shall review these requirements and any additional requirements needed for the supplied equipment shall be deemed to be included in the Contract Sum. If the Contractor modifies the Drawings, the Contractor shall submit these modification(s) to the S.O. for approval. Any approved modifications shall be at the Contractor's own cost.
- (b) The dimensions and positions of the openings and pipe penetrations shown in the Drawings and/or the Specifications are indicative only and are for the purposes of conveying to the Tenderers the indicative locations and nominal opening/penetration sizes.
 - (i) For equipment within the Contractor's scope, the Contractor is required to ascertain the actual dimensions and position of openings and pipe penetrations to suit the equipment selected.
 - (ii) The Contractor shall interface with Other contractors and the S.O. to ascertain the actual dimensions and positions of openings and pipe penetrations for the Contractor's equipment. The Contractor shall provide this information to the S.O., in a timely manner as directed by the S.O., to allow for incorporation into the design and prior to the start of construction by Other contractors. Failure to provide this information in a timely manner, shall require the Contractor to provide the penetration (with the S.O.'s approval) at his own cost.
- (c) The Contractor shall comply with Appendix P2D – Schedule of Submissions.

P2.15 INFORMATION MANAGEMENT

P2.15.1 Overview

- (a) The Contractor shall comply with the requirements for Information Management given in the General Specification G1.2 Information Management.
- (b) The delivery of TWRP requires all parties (the Board, the S.O. and contractors) to embrace digital engineering, also known as Building Information Management (BIM).

P2.15.2 Building Information Management (BIM)

- (a) The Board's Information Requirements (BIR) for BIM are described in the EIR and General Specification G1.2 Information Management.
- (b) All information, models, drawings, documentation and recording of the progress of the Works shall comply with the requirements of the EIR and General Specification G1.2 Information Management.

- (c) The Contractor shall integrate the Works with the Other contractors and with the works executed as part of other contracts. This integration will require numerous interfaces and coordination. The Released Models shall be developed to allow performance of multi-disciplinary interference checks. The Contractor shall provide expert personnel with the required skills to use the BIM Model(s) and databases to satisfy the above requirements. The Contractor shall conduct regular design reviews and clash checks with the Board and the S.O.. Two-dimensional (2D) drawings will not be accepted for this purpose.
- (d) The Contractor shall be responsible for updating the P&IDs and developing the associated databases for the entire Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities. This shall include updating the P&IDs and databases with the Other contractors' scope of work as shown in the Digestion Pre-treatment and Sidestream Nitrogen Removal P&IDs. The Other contractors will provide the information for the databases (in a format approved by the S.O.) to the Contractor. Any changes to the P&IDs with reference to the Other contractors' scope of work, shall be marked-up on the P&IDs by the Other Contractors (and approved by the S.O.) and the Contractor shall update the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities P&IDs and databases. All costs associated with updating the P&IDs and developing the databases with the Other contractors' information shall be included for in the Contract Sum.
- (e) Two-dimensional (2D) drawings for submittal to the S.O. shall be extracted from the BIM model and shall not be generated as standalone drawings. This excludes select schematic drawings for which drawings generated in 2D will be acceptable.
- (f) The Contractor shall ensure as far as is practical, that vendor-supplied equipment shall be accurately modelled in 3D by the vendors and shall be incorporated into the BIM model by the Contractor. Such items shall include, but shall not be limited to:
 - (i) Process and mechanical equipment;
 - (ii) Lifting equipment;
 - (iii) Electrical components (switchgear, transformers, MCCs, UPS etc.); and
 - (iv) Instrumentation, sampling and analysing equipment;
 - (v) Building services equipment; and
 - (vi) Architecture elements.
- (g) All cost incurred for BIM modelling shall be deemed included in the rates and prices.

P2.15.3 BIM Tender Models and Drawings

- (a) A Preamble for BIM models and Drawings is provided in Volume 5.
- (b) The BIM Tender models and Drawings prepared by the S.O. describe the design intent at the time of Tender.
- (c) The BIM Tender model and Drawings provide a reference design to allow the Contractor to appreciate the general arrangement of the plant and equipment in the interfaces and spatial requirements within the contract boundary.. However, it is the Contractor's responsibility to provide all Works to meet all Specification requirements. The tender design/layout information is not intended accurately to describe the detailed design of the equipment/systems and does not in any way absolve the Contractor from meeting his contractual obligations.

P2.15.4 Engineering Documents, Models and Plans to be Furnished by the Contractor

- (a) General
 - (i) The Clauses below describe the various engineering documents and plans to be submitted by the Contractor to the S.O..
 - (ii) The procedures for submitting and obtaining approval of them shall be in accordance with the EIR and General Specification G1.2 Information Management.
 - (iii) The Contractor shall be responsible for producing and submitting for the S.O.'s review and approval, full details of the design of the Works.
 - (iv) The Contractor shall submit drawings showing the setting out of the Works.
- (b) Submission of Documents
 - (i) The Contractor shall comply with the document control requirements in the EIR and General Specification G1.2 Information Management.
 - (ii) The Contractor shall maintain a model, drawings and specification control programme to ensure construction in accordance with the latest model/drawing revision. The drawing control programme shall also include drawing records, shop drawings, and detail drawings required for the work. Contractor's drawings are subject to review by the S.O..
 - (iii) The Contractor's documentation and models/drawings shall be subject to review by the S.O. and are to be submitted as individual files to the S.O. in accordance with the document control system and process as specified. Batch files will not be accepted.
 - (iv) The Contractor shall establish procedures to ensure compliance with requirements in the Specifications which describes but does not limit the documents to be supplied by the Contractor to complete the Work.

(c) Design

- (i) The design of the Works shall be an integral part of the Contract. Such design shall include design of temporary works, designs of permanent works, models, drawings, functional plans and general designs, together with all other studies, investigations, computations, analyses and evaluations necessary to comply with the requirements of the Specifications.
- (ii) Functional plans and designs shall be based on the proposals submitted with the Contractor's Tender, subject to such modifications as the Board and the S.O. may require under the terms of the Contract.
- (iii) All designs, models, drawings and documents relating to design of the Works shall be submitted by the Contractor to the S.O. for review and approval. Submissions shall be accompanied by corroborative information and computations as necessary to enable the S.O. to check the Contractor's designs.
- (iv) All submissions shall be in English and shall use SI units.
- (v) The Contractor shall be fully responsible for the content of all submissions to the S.O. irrespective of the source or origin of information contained in such submissions. All submissions irrespective of the source or origin of information shall be checked by the Contractor and endorsed as such before submission to the S.O. who will not accept submissions for review unless these requirements have been met.
- (vi) Review of designs, model, drawings and documents by the S.O. will mean review in principle only and not verifying every dimension and detailed matter.
- (vii) The Contractor shall carry out the Works only in accordance with designs, model, drawings and documents, which have received the S.O.'s written approval.

(d) Modelling and Drawing Standards

- (i) The Contractor shall comply with the modelling and drawings standards as set out in the EIR and General Specification G1.2 Information Management.
- (ii) All drawings and model shall be in electronic format, occasional hard copies may be required for the execution of work at Site.
- (iii) Where 2D/3D drawings are revised, the revision letter or number shall be incorporated in the title block and the revision shall be clearly indicated on the drawings with the revision letter or number shown in an adjacent triangle.

(e) Form of Drawings

- (i) Every drawing shall have a title box in the bottom right corner showing:
 - 1) Board's name and logo;
 - 2) S.O.'s name and logo;
 - 3) Title of scheme;
 - 4) Title of Contract;
 - 5) Contractor's name;
 - 6) Title of work location and area code;
 - 7) Title of drawing;
 - 8) Drawing number;
 - 9) Date;
 - 10) Author; and
 - 11) Signature of Contractor (to the effect that the drawing, whether his own or from any other source, has been checked by him before submission to the S.O..
- (ii) Each drawing shall also have a separate revision box with space for up to 10 revisions, and including revision number, revision date, revision description and revision check. It shall also include adequate clear space for the S.O.'s markings.
- (iii) Drawings shall be drawn to specified scales or to such scales as are appropriate for clearly detailing and conveying the Contractor's proposals. Scales shall generally be 1:2, 1:5, 1:10 or multiples of 10 thereof. The appropriate measuring scales used shall be shown on the drawings.
- (iv) Drawings shall include cross references where appropriate and key information such as vital levels and dimensions.

(f) Form of Non-Graphical Data (Designs and Documents)

- (i) Designs and documents submitted by the Contractor to the S.O. for approval shall comprise generally:
 - 1) Contents list;
 - 2) Scope (description of the contents and purpose of the submission);

- 3) Conclusions and recommendations;
 - 4) References, specification requirements, codes, manuals and supporting documents used, model ref, drawing numbers, and titles of drawings which are based on the design;
 - 5) Description of design approach;
 - 6) Criteria, parameters and methods used;
 - 7) Test procedures, analyses and results;
 - 8) Calculations and schedules;
 - 9) Qualitative description and comments on results;
 - 10) Any other relevant information; and
 - 11) Appendix.
- (ii) Designs and documents shall be presented with every page numbered. The title of the submission shall be given on the front sheet together with a reference number, Board's name, title of scheme, title of Contract, Contractor's name, title of works location, author's reference, date, Contractor's signature, revision box and space for S.O.'s markings as for drawings and any other relevant information.
- (g) File Naming Convention
- (i) The Contractor shall comply with the EIR and General Specification G1.2 Information Management for the file naming convention.
- (h) Submission Procedures
- (i) The procedures for submission shall be in accordance with the EIR and General Specification G1.2 Information Management.
- (ii) Every model/ drawing submitted by the Contractor to the S.O. for review and approval shall be:
- 1) Based on previously approved designs or documents. Interrelated models/drawings shall be submitted at the same time in a complete and self-sufficient set; and
 - 2) Accompanied by a letter from the Contractor (not from subcontractors or suppliers). Revisions of previous submissions shall be highlighted and the reasons for these shall be stated in the covering letter.

- (iii) All documents and drawings describing Contractor's design, and which are required for statutory submissions shall be endorsed by a Professional Engineer (PE) registered in the appropriate discipline with the Professional Engineers Board of Singapore.
- (iv) All designs and documents submitted by the Contractor to the S.O. for approval shall be checked by the Contractor before submission. The Contractor shall likewise check submissions from his subcontractors or from any other source before passing on such submissions to the S.O. for approval:
 - 1) Subject to any specific requirements in the Contract, all submissions shall be made sufficiently early to allow the S.O. time for review and approval, and for the Contractor to provide and submit revisions or alternatives.
 - 2) Unless otherwise specified or agreed before submission, the Contractor shall allow not less than twenty (20) working days excluding weekends and public holidays for review by the S.O. of designs documents and samples which are required to be approved and fourteen (14) working days for review of re-submissions before manufacture or construction commences as part of the Permanent Works. The Contractor shall have demonstrated in his submissions that he has coordinated submissions by his subcontractor or suppliers to confirm that the Contractor is satisfied that they are acceptable for incorporation into the Works.
 - 3) If the S.O. does not respond to any submission within the specified number of days, then the Contractor may proceed with the work but shall inform the S.O. in writing that he is proceeding with the work. Absence of response from the S.O. will not relieve the Contractor of his obligations to meet all the requirements of the Specifications.
 - 4) The S.O. will designate the address of his review office in respect of disciplines or work from time to time during the Contract. All submissions shall be made sufficiently early to allow the S.O. time for review and approval of model, drawings, documents and samples which are required to be approved before commencement of manufacture of plant or of installation or construction on-Site as the case may be and for the Contractor to provide and submit revisions or alternatives, or carry out further tests if, in the opinion of the S.O., the initial submissions do not meet the requirements of the Contract.

- 5) Whichever status - “Approved”, “Approved subject to” or “Not Approved” the S.O. assigns to any models, drawings, designs or documents submitted for his review, will be based on the information provided in the submitted models, drawings designs or documents. It shall be the responsibility of the Contractor to ensure that all relevant information is provided to enable the S.O. to easily determine if a submitted design complies with the specific requirements or intent of the Specifications. Where the S.O. assigns an inappropriate status due to lack of clarity or incomplete or incorrect information submitted by the Contractor, this shall not relieve the Contractor of his obligations to provide the Works in compliance with the Contract. Any costs for abortive works or redundant equipment provided because of such situations shall be borne by the Contractor.

(i) Programme of Document Plans and Submissions

- (i) The Contractor shall comply with the milestones for purposes of interfacing with Other contractors and to fulfil the TWRP completion milestones. The Contractor shall submit all specified and necessary plans and other submissions required for the planning and execution of the Works. All plans and submissions shall be included in the Contractor’s programme and be submitted to the S.O. sufficiently in advance of the works to allow for S.O. review and approval in accordance with the requirements of the Specifications including EIR and G1.2 Information Management.
- (ii) Plans shall be revised and updated as necessary during the Contract period to address the progress of the Works and changes in the Site.
- (iii) The Contractor shall submit all required plans to the S.O. expeditiously following the Commencement Date.
- 1) Project Execution Plan;
 - 2) Quality Plan;
 - 3) Safety Management System;
 - 4) Programme for the Works;
 - 5) Expenditure projections;
 - 6) BIM Execution Plan (BEP);
 - 7) Detailed Submittal Register;
 - 8) Proposal for concrete trial mix and mix design (if any); and
 - 9) Authority submission for temporary works (if any).

- (iv) The Contractor shall prepare a programme for submitting designs, models, drawings and documents to the S.O. for approval. The programme shall base on Contractor's detailed submittal register and provide for the S.O.'s review of the Contractor's submissions to be undertaken at a reasonably steady rate of working. The programme shall also make reasonable provision for re-submission of not approved designs, models, drawings and documents and for the time needed to transmit such designs, models, drawings and documents. No designs, models, drawings and documents will be accepted by the S.O. for review until the programme for their submission has been approved by him; and
- (v) The S.O. may withdraw or modify his approval to an "Approved" design drawing or document because of subsequent submissions. In this circumstance, he will inform the Contractor in writing of the withdrawal or modification and, upon receipt of such information, the Contractor shall immediately take the necessary action regarding the Works and confirm this to the S.O. in writing.
- (j) Definition of Working Days for Approval of Submissions:
 - (i) In the context of time for approval by the S.O. of submissions by the Contractor, and for no other, "working days" shall mean Monday to Friday in each week and shall exclude any public holidays in Singapore. The number of working days shall be counted from the next working day following the day of delivery to the office of the S.O. until the day before the day of despatch from the office of the S.O. to the Contractor.

P2.15.5 BIM As-Built, As-Built Drawings

- (a) At the end of the project, the Contractor shall provide as-constructed information to the S.O..
- (b) The as-constructed information shall include 'As-Built' BIM models, associated extracted 'As-Built' drawings and the associated database and shall be submitted to the S.O. in accordance with the EIR and General Specification G1.2 Information Management. This shall include the following:
 - (i) Where Model was provided for Construction, these shall be updated with accurate As-Built information which includes it not limited to:
 - 1) Actual position, element, finishes, surfaces and build-up (e.g. blockwork, insulation, plasterboard, paint for walls);
 - 2) Actual manufacturer details, installation date, classification;
 - 3) Full COBie data – as defined with S.O. upon award of the Contract; and
 - 4) Asset ID;

- (ii) Intelligent schematic drawings (such as P&IDs) shall be developed and updated in the database and drawings to include all fields required for the Board's Asset Management and actual installed items (to be provided for after the contract Commencement Date);
 - (iii) All Contractor designed elements shall be submitted as an intelligent model complete with information given in item (i) and (ii) above; and
 - (iv) All other remaining items not modelled shall be updated as 2D drawings although a 3D model equivalent will be accepted.
- (c) The As-constructed information shall include '2D As-Constructed drawings' and shall be submitted to the S.O. in accordance with the EIR and General Specification G1.2 Information Management. This shall include the following:
 - (i) The Contractor shall submit three (3) sets in A1 format and five (5) sets in A3 format of As-Constructed drawings to the S.O.;
 - (ii) The Contractor shall be responsible for provision of all necessary hardcopies or soft copies of the As-Constructed drawings for S.O. submission to BCA for application of Certificate of Statutory Completion (CSC). The Contractor shall allow for the cost of all production, resubmission, printing and scanning of the As-Constructed drawings for the application for the Certificate of Statutory Completion (CSC) and for closing out with the Board;
 - (iii) The drawings, and particularly dimensions and notes, shall be prepared so that the drawings are still legible at this size. If in the S.O.'s opinion the drawing is not legible or clear at this size, then the drawing shall be resubmitted at the appropriate size as defined by S.O.; and
 - (iv) The As-Constructed drawings shall be submitted in accordance with the EIR and General Specification G1.2 Information Management.

P2.15.6 Other Submissions

- (a) Correspondence
 - (i) All formal correspondence shall be delivered via the TWRP PMIS as detailed in the EIR and General Specification G1.2 Information Management.
- (b) Test Reports
 - (i) In addition to any specific obligations for sampling and testing, the Contractor shall be responsible for routine inspection sampling and testing of all materials, workmanship, and plant and measuring devices, to control the quality of work and to ensure compliance with the Specifications and with approved samples.

- (ii) Any consumables used by the Contractor in furtherance of testing specified in the Contract, or which he may require for his own purposes shall be provided by the Contractor at his own expense.
- (c) Samples
 - (i) Where the Contract requires the submission of samples, they shall be submitted by and at the expense of the Contractor not less than 20 working days excluding weekends and public holidays prior to the time that the materials represented by such samples are needed for incorporation into any work. Samples shall be subject to approval by the S.O., and material represented by such samples shall not be manufactured, delivered to the Site nor incorporated into any work without such approval.
 - (ii) Where samples, including samples of materials and workmanship constructed on the Site, are submitted as a reference for materials and workmanship to be provided as part of the Permanent Works, they shall, after being approved by the S.O., be carefully preserved for this purpose by the Contractor to the satisfaction of the S.O. until permission is given by the S.O. for their disposal.
- (d) Manufacturers' and Contractor's Certificates
 - (i) Where certificates are required by the Specifications or relevant Reference Standard, the original and one copy of each such certificate shall be provided by the Contractor, unless otherwise specified. The original and copy certificates shall be submitted to the S.O..
 - (ii) Certificates shall be clearly identified by serial or reference number where possible to the material being certified and shall include information required by the relevant Reference Standard or Specifications clause.
 - (iii) Unless otherwise specified, the timing for submittal of certificates shall be as follows:
 - 1) Manufacturer's test certificates shall be submitted as soon as the tests have been completed and, in any case not less than seven calendar days prior to the time that the materials represented by such certificates are needed for incorporation into the Permanent Works; and
 - 2) Certificates of tests carried out during the installation of plant and equipment or on completion of parts of the Permanent Works shall be submitted within seven (7) days of the completion of the test.

P2.16 PROTECTION AGAINST CYBERSECURITY THREATS

- (a) The Contractor shall protect the Board and the S.O. against cybersecurity threats.
- (b) The Contractor shall ensure that all equipment and devices used for the development and transfer of information to the Board and S.O. be free from virus and malware. At a minimum, the Contractor shall carry out the following:
 - (i) All computers shall be installed with licensed and recognised anti-virus software and updated with the latest definition. These shall be scanned for malware using the licensed anti-virus software before carrying out any works and show proof of the scan as requested by the S.O..
 - (ii) All storage / removable devices used on the project shall be scanned for malware before carrying out any works and show proof of the scan as requested by the S.O..
 - (iii) In the event of a cybersecurity incident, the Contractor shall inform S.O., and assist in the investigation. The Contractor shall be responsible to identify the threat and remove the threat to the satisfaction of the S.O..
- (c) The Contractor's responsibilities for cybersecurity protection shall include any suppliers or subcontractors engaged for the Works.

P2.17 PLANT TESTING AND COMMISSIONING

P2.17.1 Overview

- (a) The implementation of the testing and commissioning requirements shall be undertaken by the Contractor's teams as specified in the General Specification G2.26 as applicable to the Contract and the Particular Specification P10 Plant Performance Testing.
- (b) The sequence and timing of plant testing and commissioning is to be determined by the Contractor to meet the Contract key project milestones.
- (c) All costs associated with plant testing and commissioning including labour, consumables, water, pumping of seed sludge, waste removal, temporary testing equipment and instrumentation and sampling and testing are to be included in the Contract Sum.
- (d) Testing as specified in G2.26 of the Contractor's scope of works shall be clearly identified in the Testing and Commissioning plan and in the Programme of Works to meet the Contract key project milestones.
- (e) The Contractor shall be required to work with the S.O. and Other contractors for the commissioning of the Contractor's MEICA components. This includes all testing and commissioning activities associated with the Contractor's scope of Works.

- (f) It also has to demonstrate that each process system unit, component and sub system is capable of performing properly, both individually and as a whole complete Plant system.
- (g) The Contractor shall indicate clearly in the Testing and Commissioning Plan the volume of water (e.g. influent, G2D, NEWater or potable water) that it plans to use for the testing and commissioning of each system/ sub-system if it is has not been specified in the Specifications, and is subject to the approval of the S.O..

P2.17.2 Notification by Contractor

- (a) Following the successful completion of Stage 2 testing (Stage 2a and Stage 2b) of Digestion Pre-treatment, the Contractor shall seek S.O.'s approval in writing, giving 2 weeks' notice, to receive dewatered cake from pre-dewatering centrifuges and silos for preparation of the subsequent process commissioning (Stage 3a).
- (b) Following the successful completion of Stage 2 testing (Stage 2a and Stage 2b) of Sidestream Nitrogen removal system, the Contractor shall seek S.O.'s approval in writing, giving 2 weeks' notice, to receive dewatering centrate for preparation of the subsequent process commissioning (Stage 3a).

P2.17.3 Temporary Pumping and Piping

- (a) The Contractor shall provide all temporary pumping and piping required for all testing and commissioning activities.

P2.17.4 Phased Influent

- (a) During Stage 3a - Process commissioning, the Contractor shall coordinate with Other contractors and the S.O. on the:
 - (i) Quantity of Waste Activated Sludge (WAS) and/or Thickened PSD available during Stage 3a for pre-dewatering and thermal hydrolysis. The sludge shall be diverted gradually (to be confirmed by the S.O.) to allow acclimatisation of the anaerobic digesters.
 - (ii) Quantity of dewatering centrate to be treated by the Sidestream Nitrogen Removal Facility and the availability of the dilution stream during Stage 3a for Sidestream Nitrogen removal deammonification process. To allow for the progressive biomass development, the commissioning of the SSN can be phased to achieve the capacity required in time for the Plant Performance Test Period (PPTP).

- (b) If the amount of dewatered cake feed or dewatering centrate is insufficient to allow the commissioning of all the units to be carried out simultaneously, the S.O. may at his absolute discretion, agree with the Contractor the sustainable feed flow to be used during the commissioning process and subsequent Plant Performance Test Period (PPTP) and the appropriate number of process units which can be reasonably commissioned and tested under the prevailing flow conditions. Under such circumstances, the Contractor shall work with the S.O. to develop an appropriate commissioning methodology to allow the facilities to be commissioned and to demonstrate the equipment performance guarantees with the agreed maximum number of units operating.

P2.17.5 Digestion Pre-treatment Commissioning

- (a) Representatives from the Thermal Hydrolysis System Manufacturer shall be present on site during the installation, testing and commissioning phases to ensure the systems are installed, configured, tested and optimised in accordance with the system specific requirements. The Contractor shall provide Inspection Test Plans (ITPs) and other documentation signed off by the Supplier / Manufacturer to confirm their approval.

P2.17.6 Sidestream Nitrogen Removal Commissioning

- (a) Representatives from the Sidestream Nitrogen Removal Supplier shall be present on site during the installation, testing and commissioning phases to ensure the systems are installed, configured, tested and optimised in accordance with the system specific requirements. The Contractor shall provide Inspection Test Plans (ITPs) and other documentation signed off by the Supplier / Manufacturer to confirm their approval.
- (b) The Contractor shall submit the proposed seed sludge quantities, seeding, optimisation and phased ramp up of capacity in the Testing and Commissioning Plan for S.O.'s approval.
- (c) All costs associated with seed sludge for the phased commissioning are to be included in the Contract Sum.

P2.18 OPERATION AND MAINTENANCE SERVICES

- (a) The Operation and Maintenance Services shall be provided by the Contractor under a Goods and Service Contract for Part 2: Services and as specified in the Particular Specification P12 Operations and Maintenance.

P2.19 TRAINING FOR THE BOARD'S PERSONNEL

- (a) Training of the Board's personnel shall be provided in accordance with General Specification G2.27 Training Advisory and Technical Support. Training shall be provided for all Plant, equipment and systems provided under the Contract.

- (b) As the Board's operations and maintenance personnel will be mobilised to the site progressively, the Contractor shall allow for training of the Board prior to completion of the Works. Repeated training sessions prior to the completion of the Part 2 Services shall be provided as per Particular Specification P12 Operation and Maintenance (OM).

P2.20 SPARE PARTS, SPECIAL TOOLS AND TEST EQUIPMENT

- (a) The Contractor shall provide critical spare parts, special tools and test equipment for all Plant/equipment. The Contractor shall list out the critical spare parts, special tools and test equipment in Schedule of Prices (SOP) - Spare Parts, Special Tools and Consumable Items.
- (b) The Contractor shall provide additional spare parts, special tool and test equipment as described in the Schedule of Prices (SOP) - Spare Parts, Special Tools and Consumable Items.
- (c) The Board reserves the right to order spares in accordance with the recommendations of the Contractor, or to vary the list of items to be ordered, or not to order spare parts. The S.O. will advise the Contractor in this regard in a timely manner.

P2.21 APPENDICES

The appendices listed below form part of this Specifications:

- (a) APPENDIX P2A – Project Breakdown Structure
- (b) APPENDIX P2B – Interfaces with Other contractors (includes Interface Tie-In Register)
- (c) APPENDIX P2C – Employer Information Requirements (EIR)
- (d) APPENDIX P2D – Contractor's Design and Build Obligations
- (e) APPENDIX P2E – Schedule of Submissions

END OF SECTION